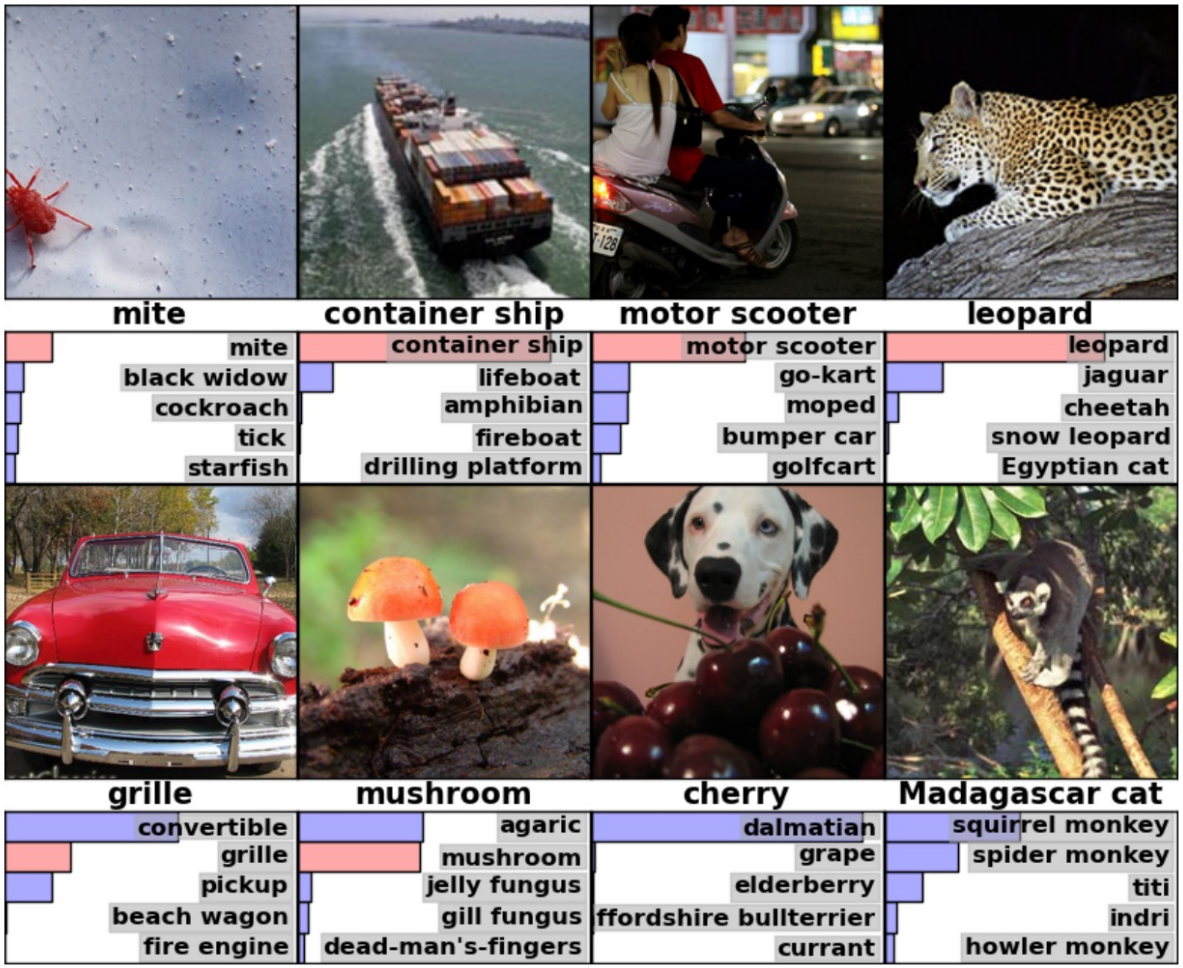
BMI / CS 771: Learning Based Methods for Computer Vision

Yin Li
UW Madison

What is this course about?

Image Classification



"ImageNet Classification with Deep Convolutional Neural Networks",
Krizhevsky, Sutskever, Hinton*, NIPS,* 2012

Object Detection and Segmentation



"Mask RCNN", He, Gkioxari, Dollár, Girshick, *ICCV,* 2017

Human Pose Estimation



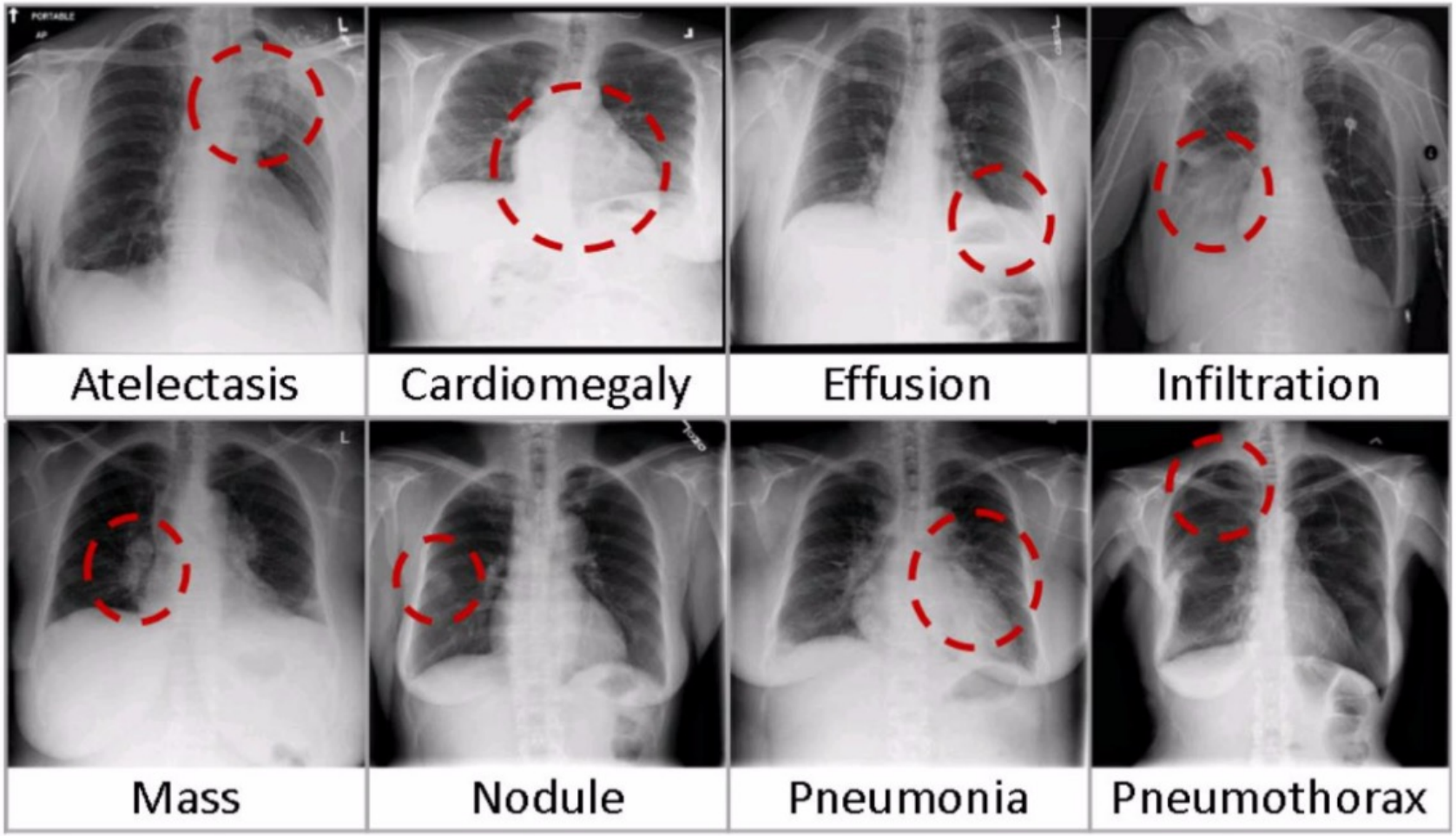
“OpenPose: realtime multi-person 2D pose estimation using Part Affinity Fields”,
Cao, Hidalgo, Simon, Wei and Sheikh, ArXiv 2018

3D Understanding (Pose and Shape)



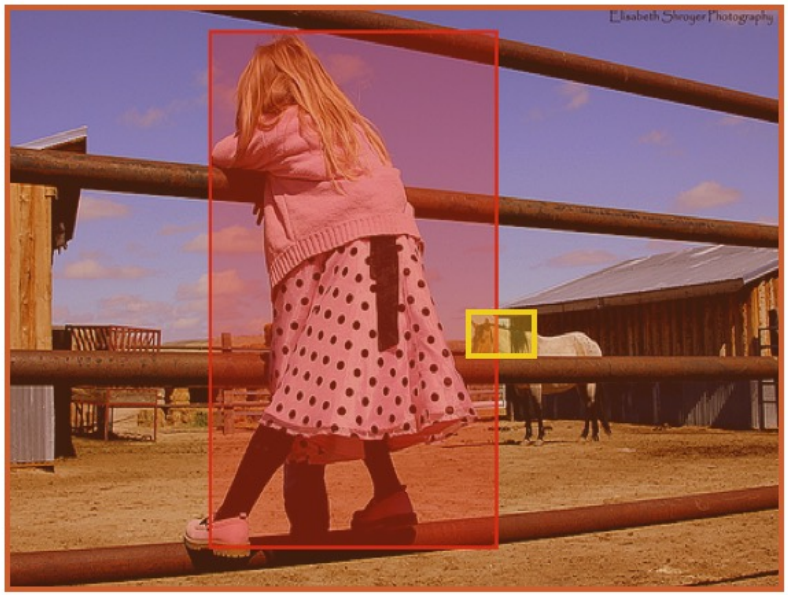
“3D-RCNN: Instance-level 3D Object Reconstruction via Render-and-Compare”,
Kundu, Li, Rehg. CVPR, 2018

Medical Image Analysis

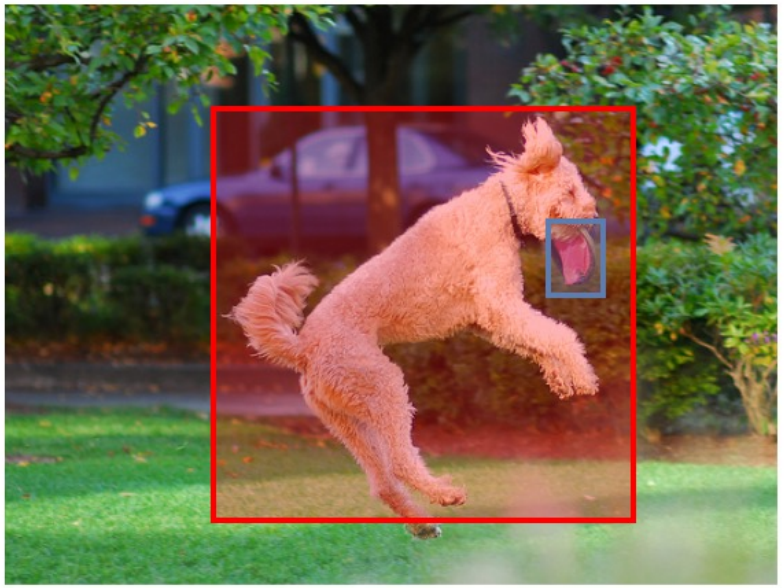


"ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases", Wang et al.*, CVPR,* 2017

Image-text Matching



A little girl stands on the fence while peeking through it to look at the horse



A large yellow dog leaps into the air to catch his Frisbee.

“Learning Deep Structure-Preserving Image-Text Embeddings”,
Wang, Li, Lazebnik. CVPR 2016.

Visual Question Answering

Who is wearing glasses?

man



woman

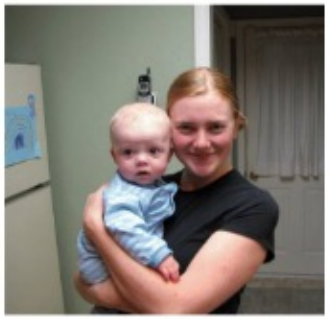


Where is the child sitting?

fridge



arms



Is the umbrella upside down?

yes



no



How many children are in the bed?

2



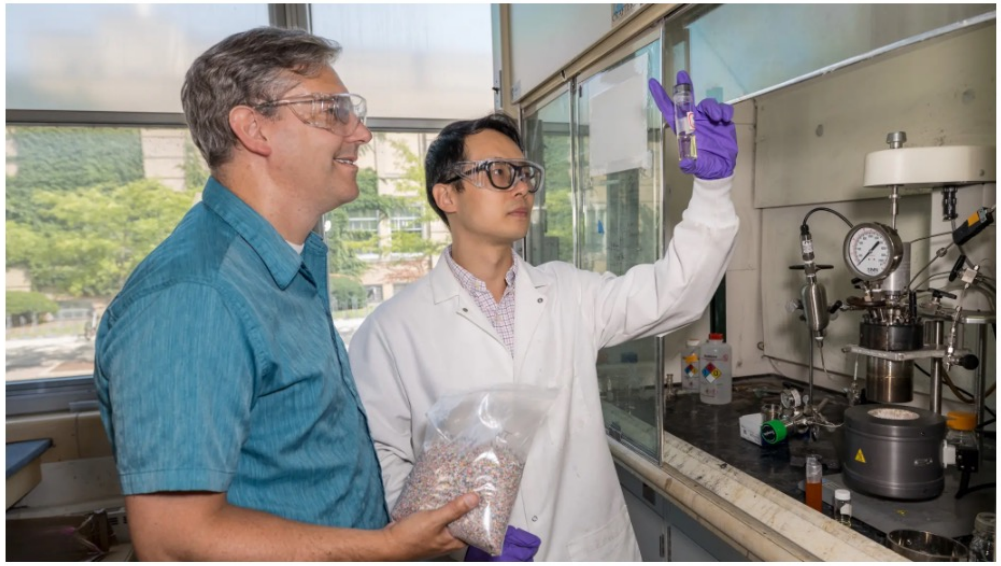
1



"Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering", Yash Goyal, Khot, Summers-Stay, Batra, Parik*, CVPR,* 2017

Multimodal Large Language Model

You



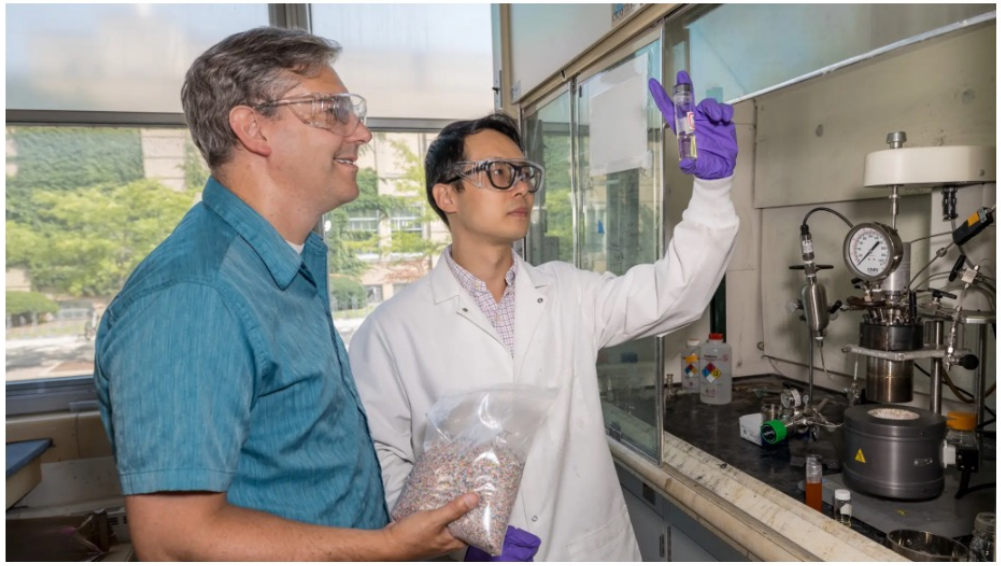
Can you describe what is in this image?

ChatGPT

The image shows two individuals in a laboratory setting. On the left, there's a man wearing glasses and a blue, short-sleeved, button-up shirt. He's holding a clear plastic bag filled with what appears to be colorful bead-like particles. On the right, another man, possibly of East Asian descent, is wearing a white lab coat, safety glasses, and purple gloves. He's holding up a vial containing a purple substance. They are both looking at the vial, with the man in the lab coat pointing at it, possibly explaining something about the substance. In the background, there's lab equipment, including what looks like a pressure gauge and other devices, suggesting they may be conducting some type of scientific experiment or analy

Multimodal Large Language Model

You



You

What is the color of the gloves worn by the person on the left?

ChatGPT

The individual on the left is not wearing gloves; their hands are bare. It's the person on the right who is wearing purple gloves.

Text-Conditional Image Generation

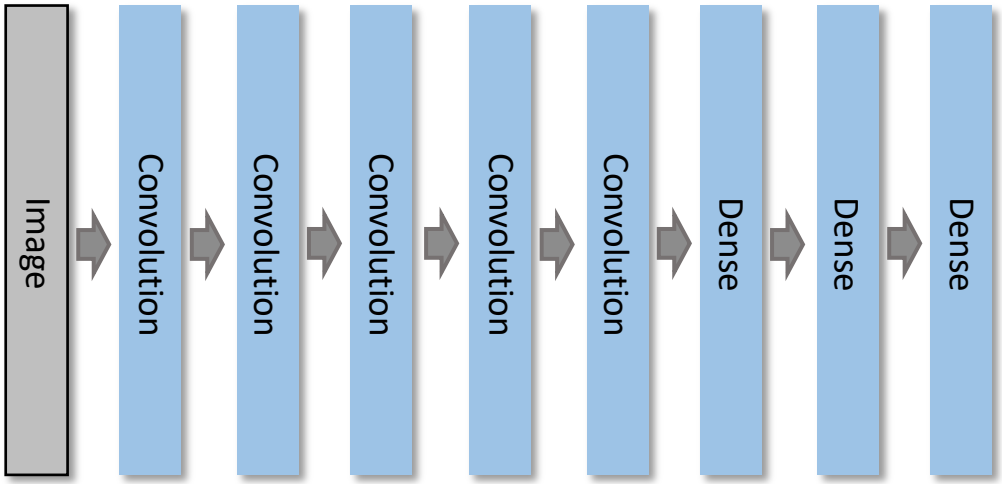
A sadhu man in Rishikesh, India meditating near the Ganges river



“High-resolution image synthesis with latent diffusion models”, Rombach et al., *CVPR*, 2023

Further reading: [How this AI image won a major photography competition]

Building deep learning models for visual recognition



How is this course organized?

https://yinli-vision.github.io/teachings/BMI-CS-771-2026/

A Peek at the Syllabus

Schedule

Week	Date	Topic	Materials
1	Wed, Sep 2	**Course Introduction / Introduction to Visual Recognition**	• Sign up for Piazza • Slides (see Canvas)
2	Mon, Sep 7	**No Class (Labor Day)**	
2	Wed, Sep 9	**Data Driven Paradigm for Computer Vision**	• Paper 1 • Paper 2
3	Mon, Sep 14	**Introduction to Neural Networks (Part I)**	• Ch 6 Goodfellow et al. • HW1 out on Canvas
3	Wed, Sep 16	**Introduction to Neural Networks (Part II)**	• Ch 6 Goodfellow et al.
4	Mon, Sep 21	**Tutorials on PyTorch and Cloud (Optional)**	• PyTorch Tutorial
4	Wed, Sep 28	**Convolutional Neural Networks (Part I)**	• Ch 9 Goodfellow et al. • HW1 due
5	Mon, Sep 28	**Convolutional Neural Networks (Part II)**	• Paper 1 • Paper 2 • HW2 out on Canvas
5	Wed, Sep 30	**Recurrent Neural Networks**	• Ch 10 Goodfellow et al.
6	Mon, Oct 5	**Transformers (Part I)**	• Paper 1
6	Wed, Oct 7	**Transformers (Part II)**	• Paper 1
7	Mon, Oct 12	**Advanced Training (Part I)**	• Ch 8 Goodfellow et al.

A Peek at the Syllabus

- Part I: Deep Learning Basics
- Part II: Classic Computer Vision Problems
- Part III: Vision Language Models
- We are still revising the syllabus. Stay tuned!

Course Information

- Instructor: Yin Li (yin.li@wisc.edu)
- TA: Khoi Duc Nguyen (kdnguyen7@wisc.edu)
- **Canvas** for announcements, homework / project submissions, slide release, etc.
- **Piazza** for online discussions
- **Website:**
  https://yinli-vision.github.io/teachings/BMI-CS-771-2026/

Course Information

- Time: Monday / Wednesday 2:30PM - 3:45PM
- Location: 3345 Engineering Hall (in-person)
- Instructor Office Hours (starting from Sep 8th):
  - Mon / Tue 11:00 am – 12:30 pm
  - In-person (Morgridge Hall Rm 6538) or over Zoom (appointment required)
  - *Feel free to email me to schedule an appointment outside normal office hours*
- TA Office Hours: TBD

Suggested Prerequisites

- Some knowledge about computer vision or machine learning (such as CS 540 / 566 / 760 / 762 / 766) or medical image analysis (such as BMI/CS 567 / 767)

- Proficient in Python programming

- This is more on *computer vision* and less on machine learning, although we will cover many aspects of deep learning

Textbook and Reading Materials

- **(Optional)** *Deep Learning* by Ian Goodfellow, Yoshua Bengio, Aaron Courville
- **(Optional)** *Computer Vision: Algorithms and Applications* by Rick Szeliski
- Free electronic copy available on their websites
- Reading materials listed on course website

Workload & Rubric

- 4 homework assignments (48% with 12% each)
  - HW1: Image processing using Python
  - HW2: Classification, Segmentation, and Detection (team-based)
  - HW3: Conditional Image Generation (team-based)
  - HW4: Multimodal Large Language Models (team-based)
- 1 final project (40%)
  - Team based (2-3 people)
  - Multiple milestones (proposal, mid-term update, final report and presentation)
- Single page course write-up (10%)
- Piazza participation (2%)

Late Submission

- Each student has three “late days” (72 lates hours) for the whole course
- Late submissions will be deducted from these late hours (and rounded up to the next hour)
- *You will have to manage your late days*

Use of AI

- Use of AI models (e.g., ChatGPT, Claude) is allowed.
- We encourage innovative use of AI tools.
- You must acknowledge its use in the writing.
- Coding agents can likely solve many of our assignments, but they cannot replace critical thinking or the learning process. Use them thoughtfully.

Before You Attend a Lecture …

- Go through the reading materials
- Lecture slides will be released before the lecture and updated the same day of the lecture. You are not required to go through the slides ☺
- Get prepared for in-class discussions

How to Succeed in the Course

- Attend the class (we don’t support remote learning)
- Cover the reading materials before each lecture
- Participate in-class and online discussions
- Teamwork for assignments and projects
- Use AI with caution!



Introduction to Visual Recognition

Many Slides from A. Efros, L. Zitnick, and A. Gupta

Outline

The challenge of visual recognition

What is visual recognition, why it is hard!



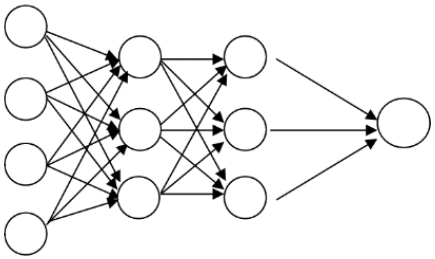
History of visual recognition

Why it works, why it fails!



Deep models for recognition

Will deep neural networks solve it all?



The Vision Story Begins

“What does it mean, to see? The plain man's answer (and Aristotle's, too) would be, to know what is where by looking.”

-- David Marr, *Vision (1982)*

Two Sides of Vision



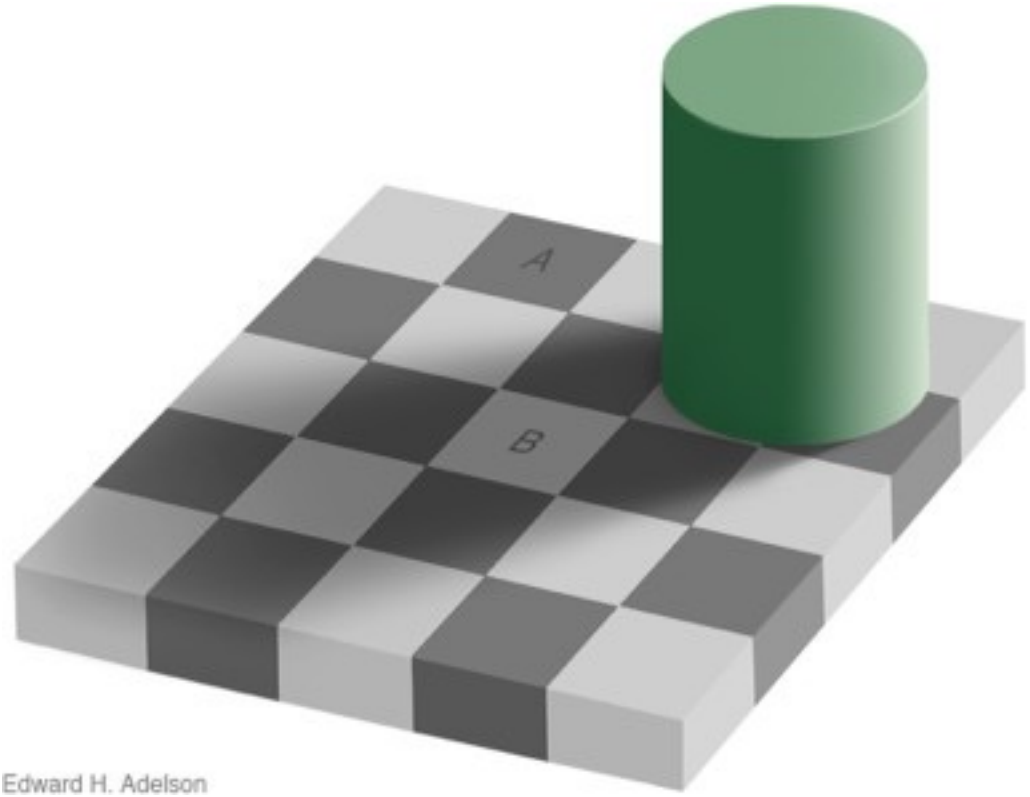
Which one do you want?

Answer #2: looks like flat sittable surface of the couch

Answer #1: pixel of brightness 243 at position (124,54) … and depth .7 meters

Slide Credit: Abhinav

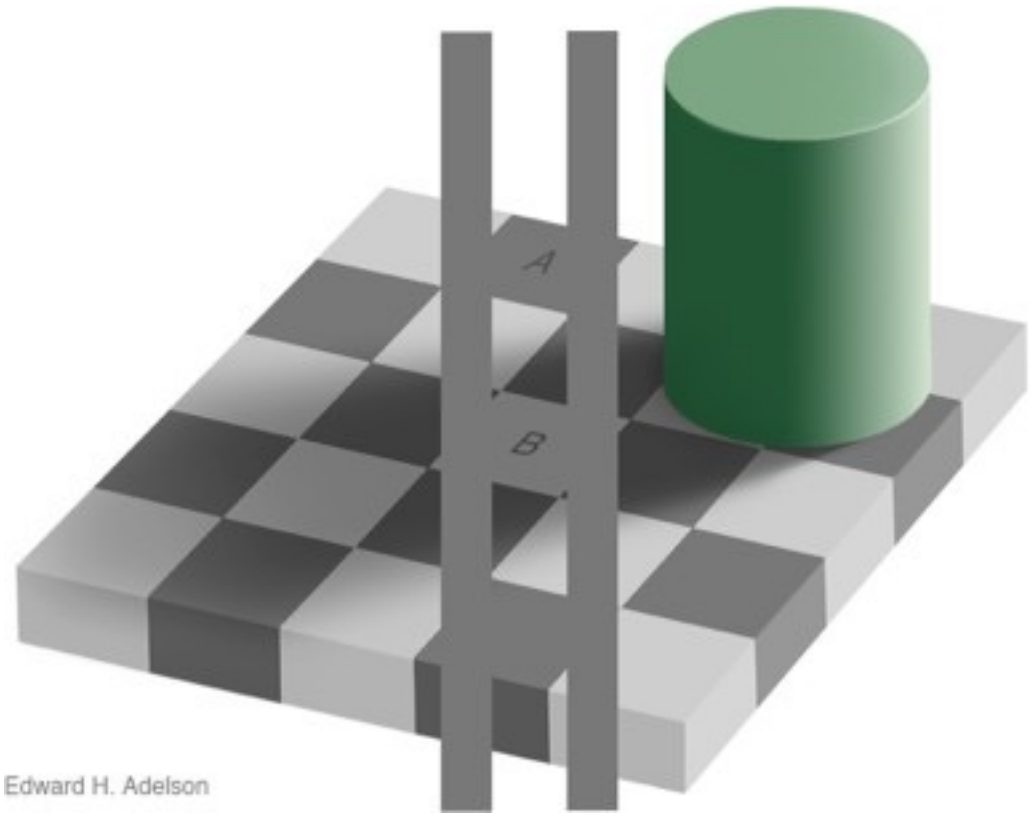
Measurement vs. Perception



Edward H. Adelson

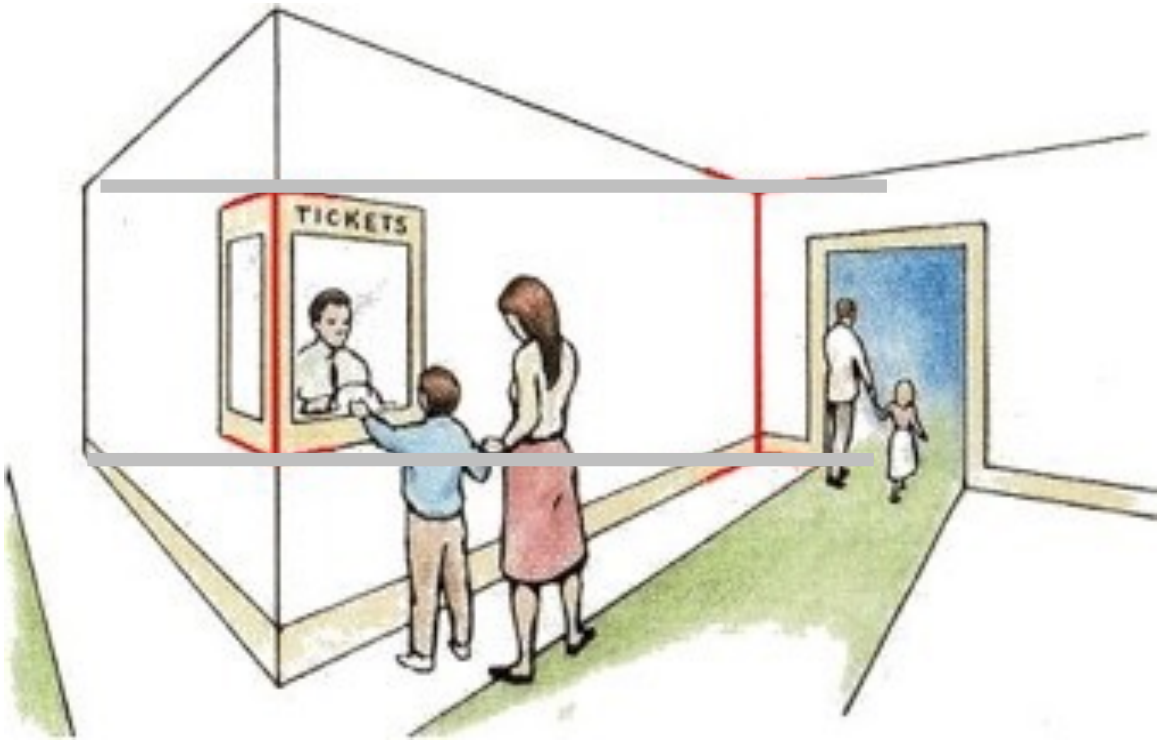
Slide Credit: Alyosha

Measurement vs. Perception



Slide Credit: Alyosha

Measurement vs. Perception



Müller-Lyer Illusion

http://www.michaelbach.de/ot/sze_muelue/index.html

Slide Credit: Alyosha

Measurement

Capturing physical quantities like pixel brightness, depth, etc.

Perception

- a high-level representation that captures the semantic structure of the scene and objects
- can be subjective – depends on task and agent
- intersection of what you see and what you infer

Visual Recognition = Perception

To infer what human perceive (what + where)

Why are the problems hard?

To infer what human perceive from 2D pixels

Our Visual World is Complex

- Viewpoint variation
- Occlusion
- Scale
- Deformation
- Background clutter
- Intra class variation
- …



Our Visual World is Complex

- Viewpoint variation



Our Visual World is Complex

- Viewpoint variation
- Occlusion



Our Visual World is Complex

- Viewpoint variation
- Occlusion
- Scale



Our Visual World is Complex

- Viewpoint variation
- Occlusion
- Scale
- Deformation



Our Visual World is Complex

- Viewpoint variation
- Occlusion
- Scale
- Deformation
- Background clutter



Our Visual World is Complex

- Viewpoint variation
- Occlusion
- Scale
- Deformation
- Background clutter
- Intra class variation
- …

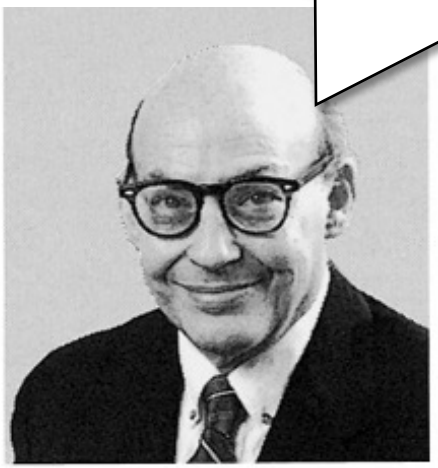


How did we approach the problems?

A brief history of visual recognition

1966

"Connect a television camera to a computer and get the machine to describe what it sees."



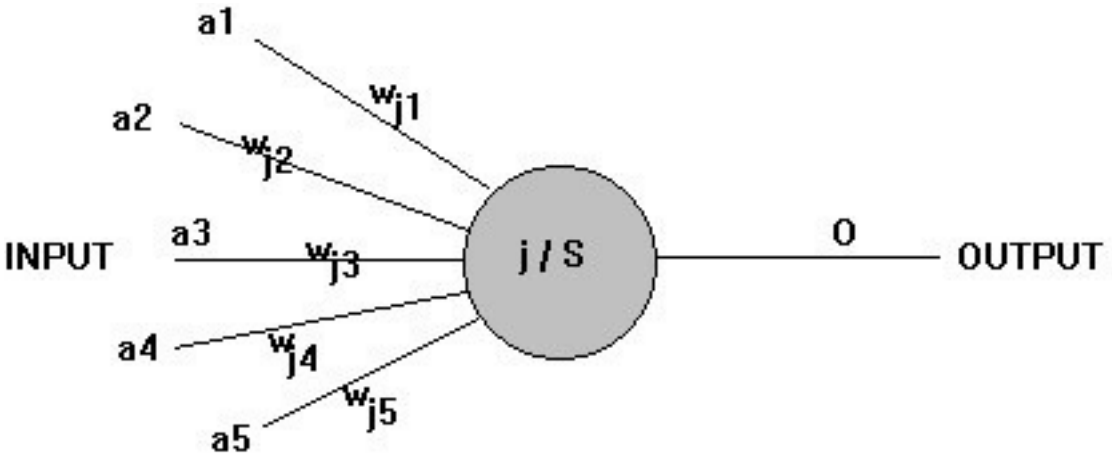
Marvin Minsky



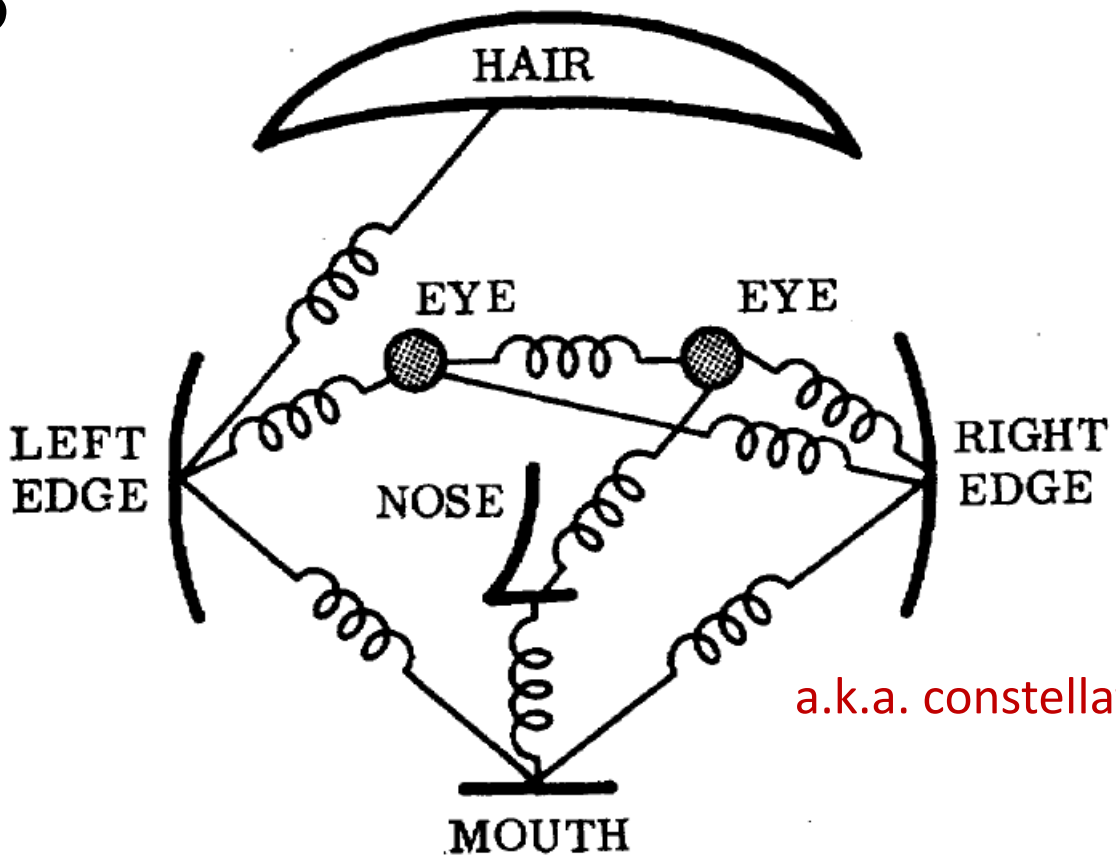
Image Credit: Wisconsin State Journal

1940’s - 1960’s

The invention and development of neural networks



1973



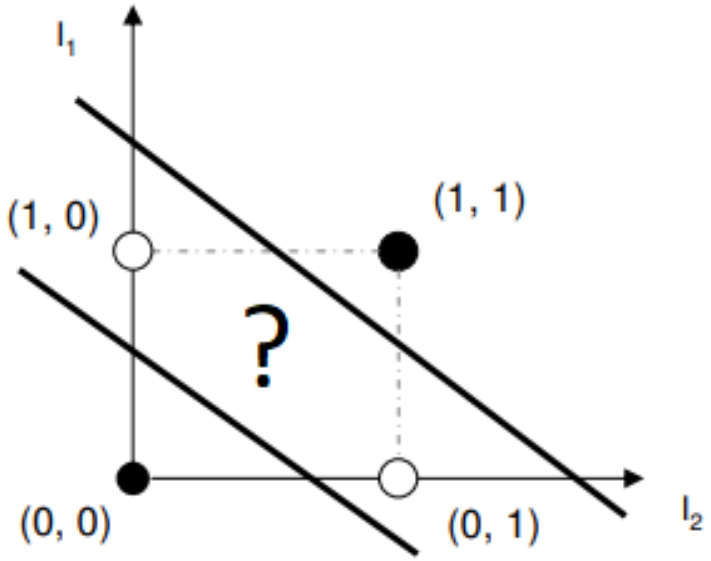
a.k.a. constellation model

The representation and matching of pictorial structures, Fischler and Elschlager, 1973

1970's

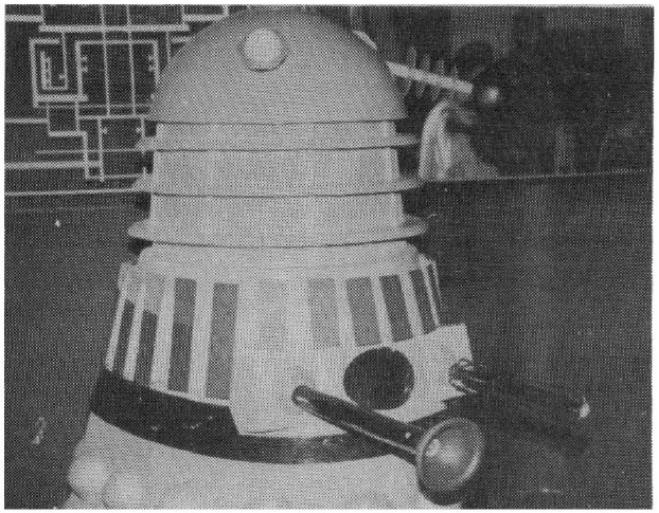
A single layer neural network can not solve XOR problem …

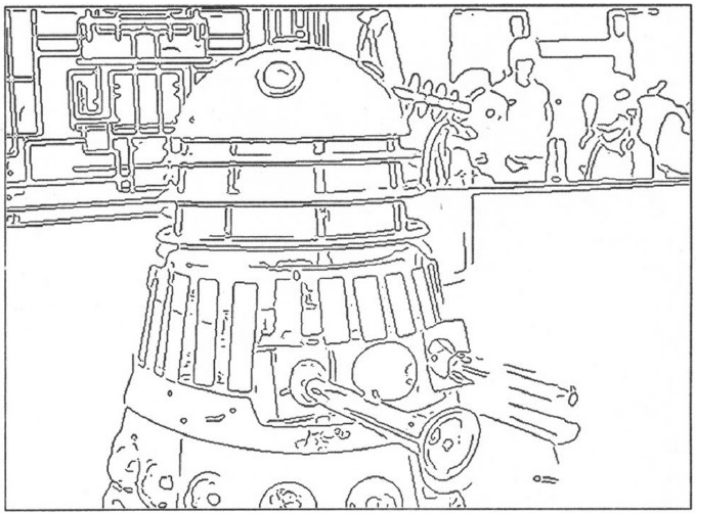
XOR		
I_1	I_2	out
0	0	0
0	1	1
1	0	1
1	1	0



1980's

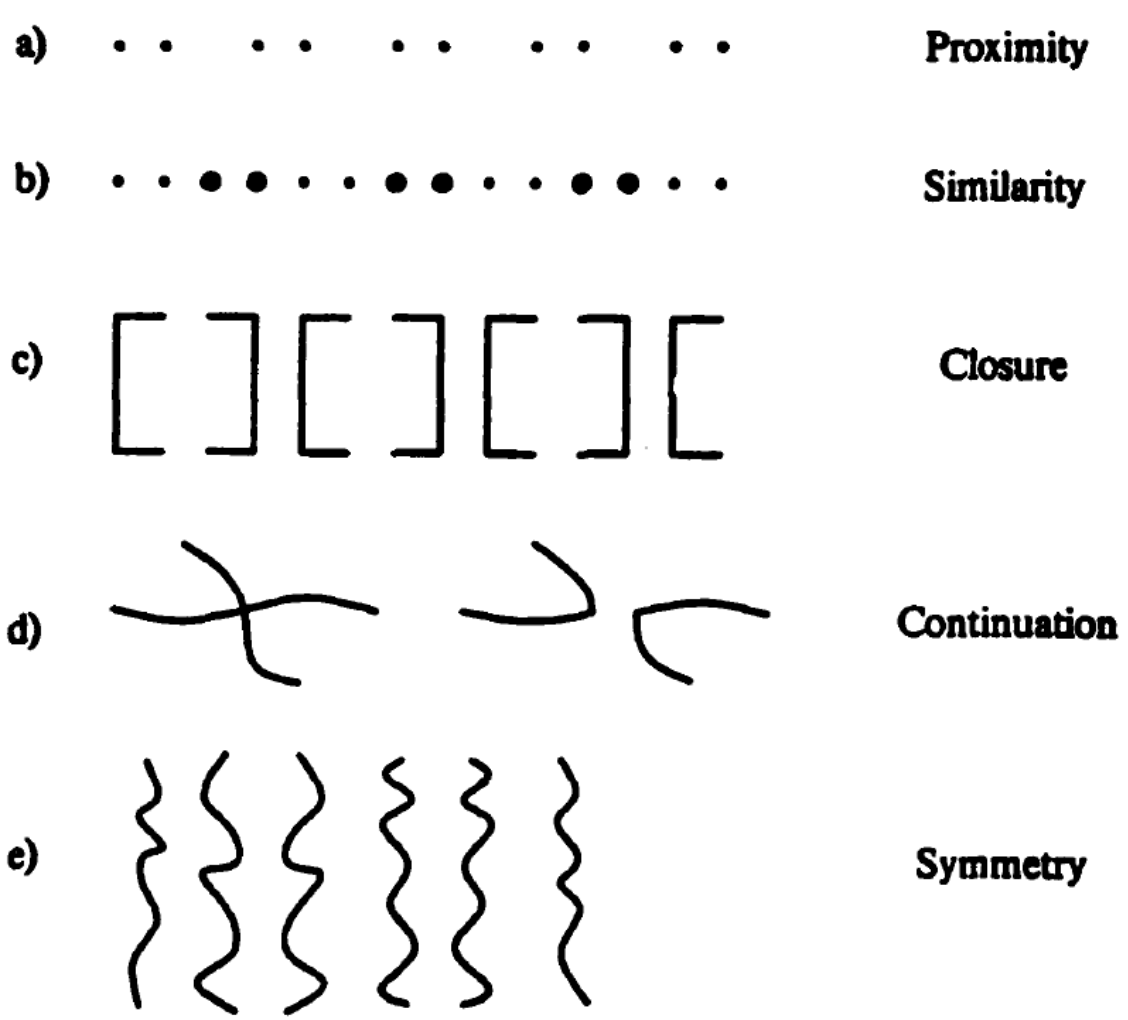
AI winter… …back to basics





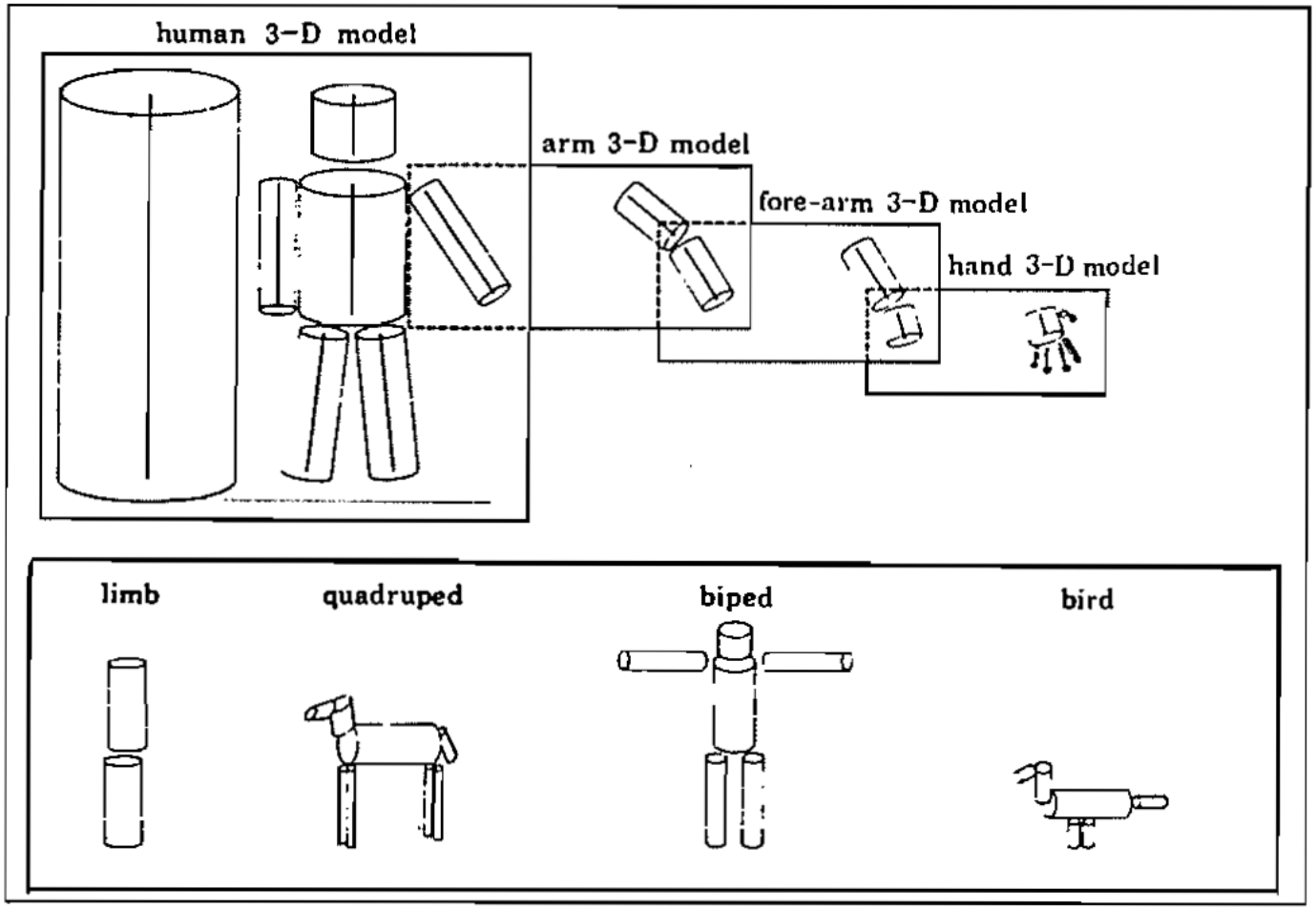
A Computational Approach to Edge Detection, Canny 1986

1984

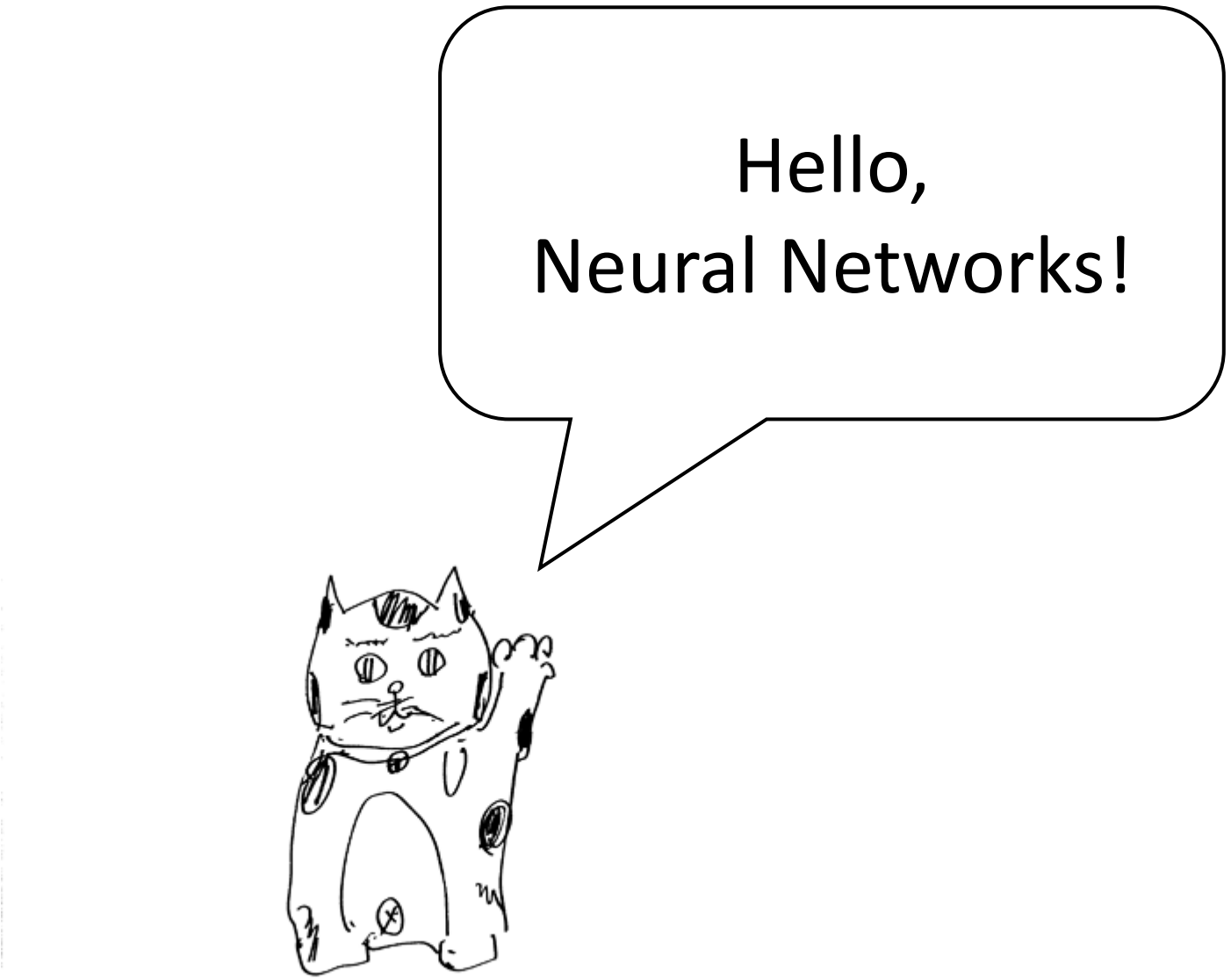


Perceptual Organization and Visual Recognition,
David Lowe, 1984

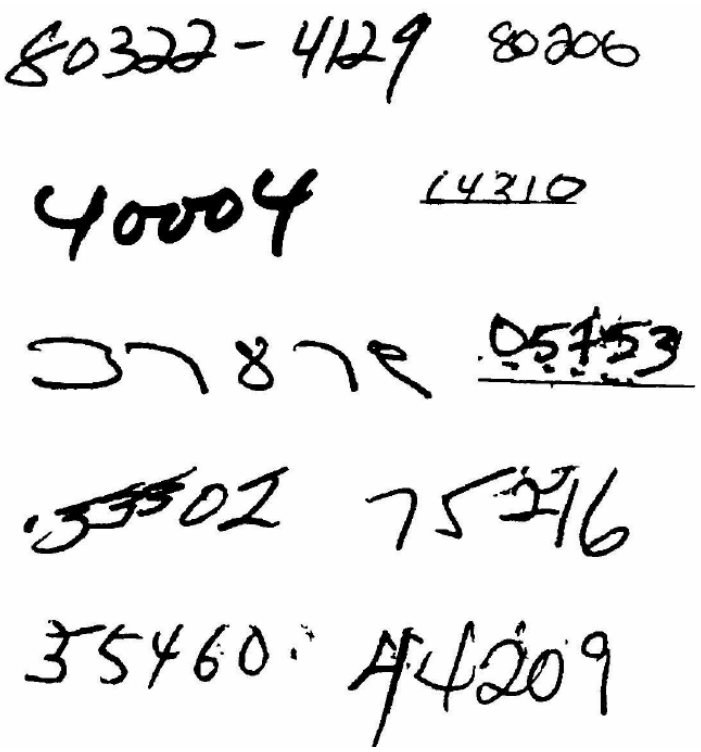
1986



Perceptual organization and the representation of natural form, Alex Pentland, 1986


Hello,
Neural Networks!

1989-1998



Zip codes

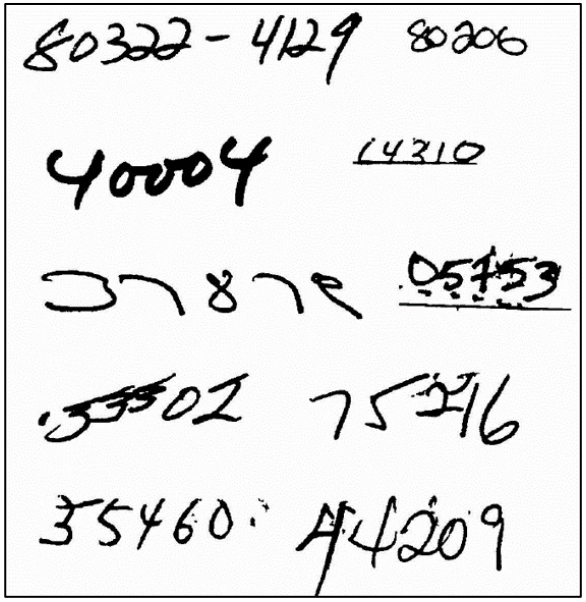
MNIST



Backpropagation applied to handwritten zip code recognition, Lecun et al., 1989

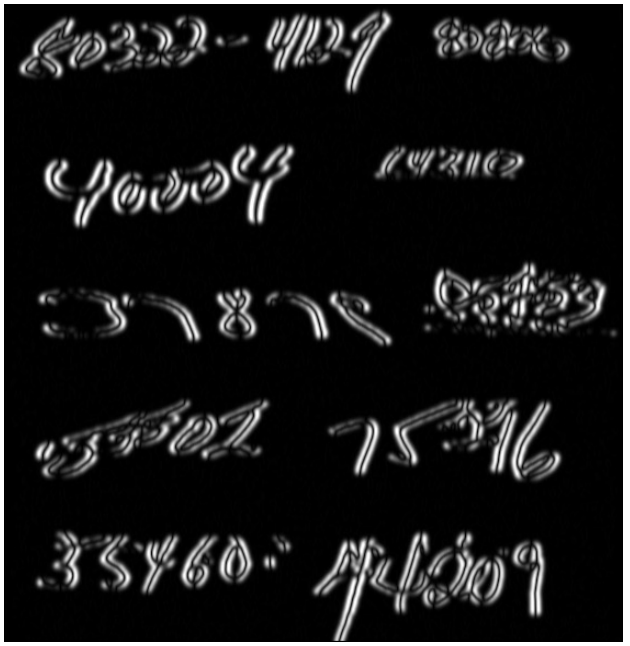
Filters (Convolutions)

Input



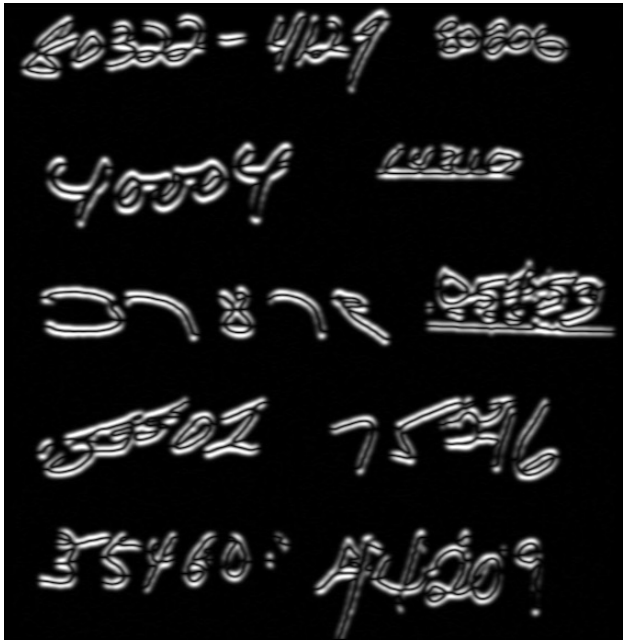
-1	0	+1
-2	0	+2
-1	0	+1

x filter

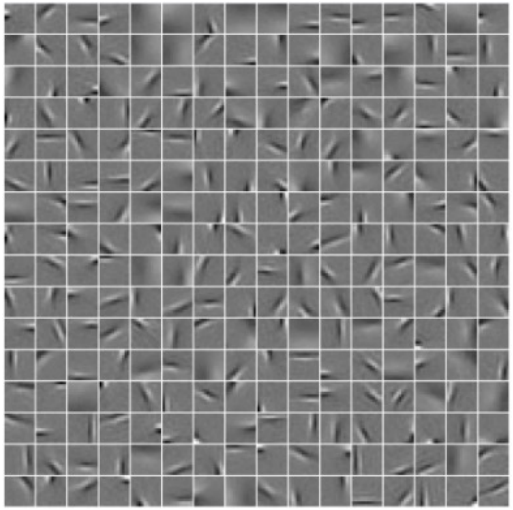


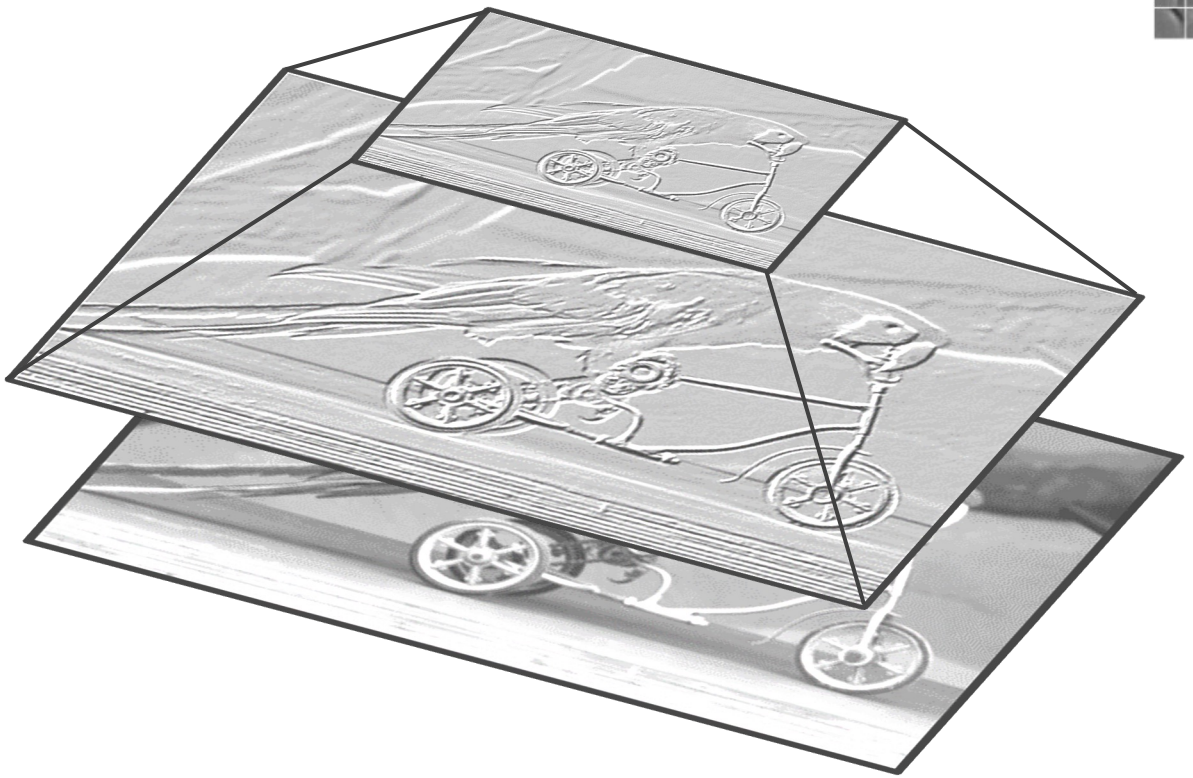
+1	+2	+1
0	0	0
-1	-2	-1

y filter



1989



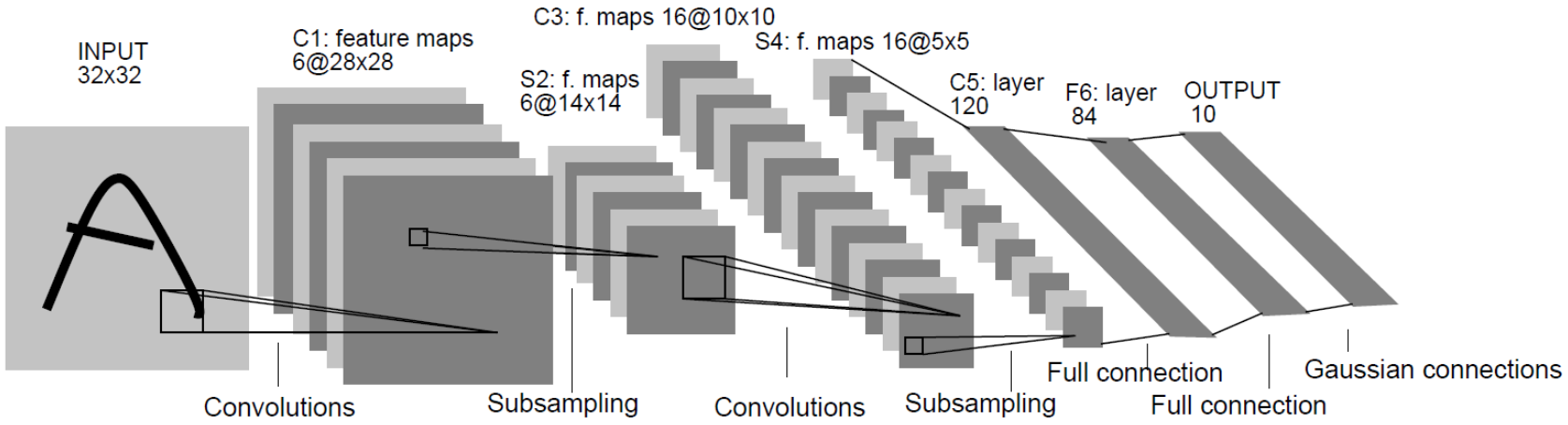


Pooling

Convolution
+ Nonlinear function

Image

1989



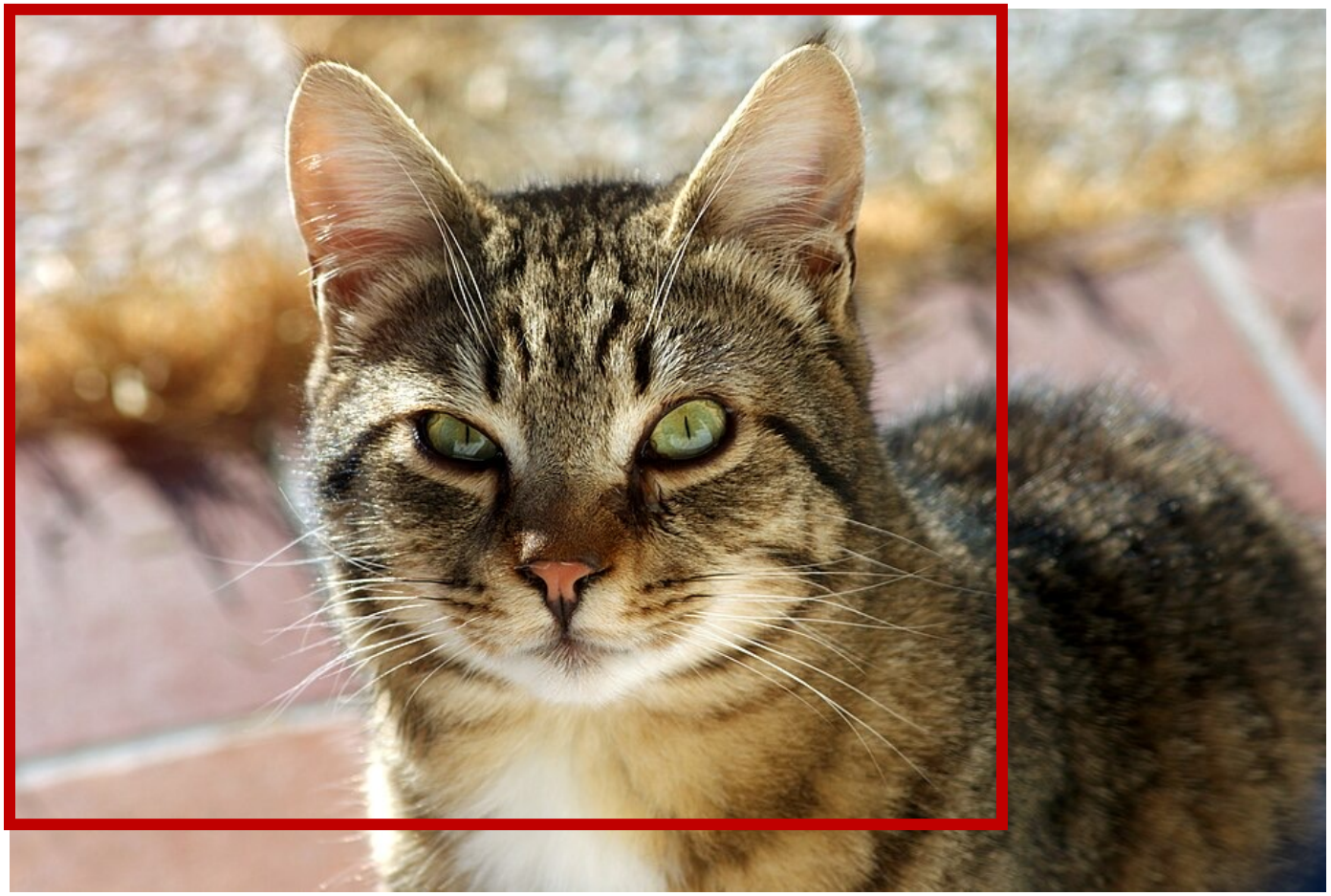
All filter weights are learned from data

Backpropagation applied to handwritten zip code recognition,
Lecun et al., 1989

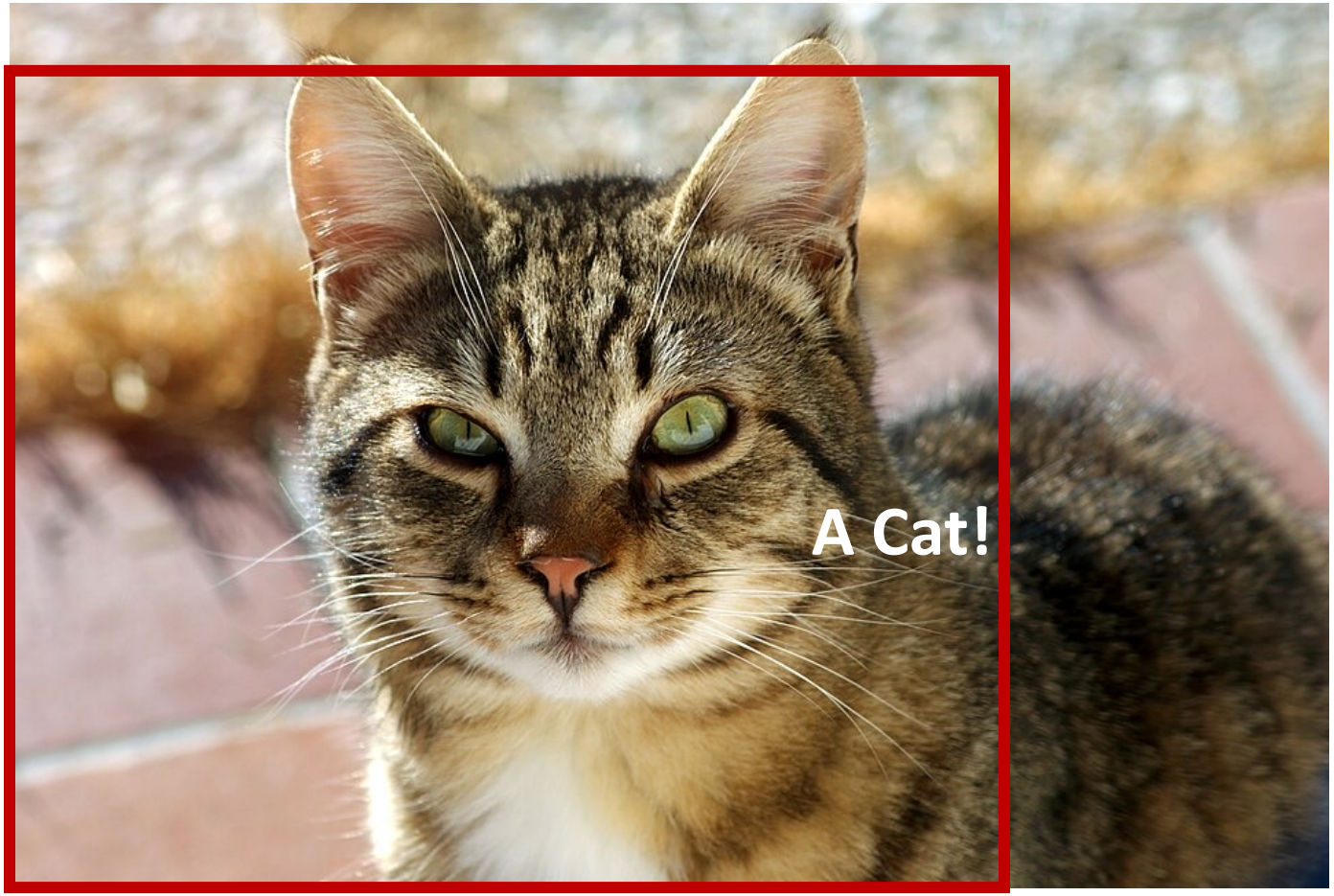


1998-2021: Detecting Faces

Sliding windows
& Classifier each window

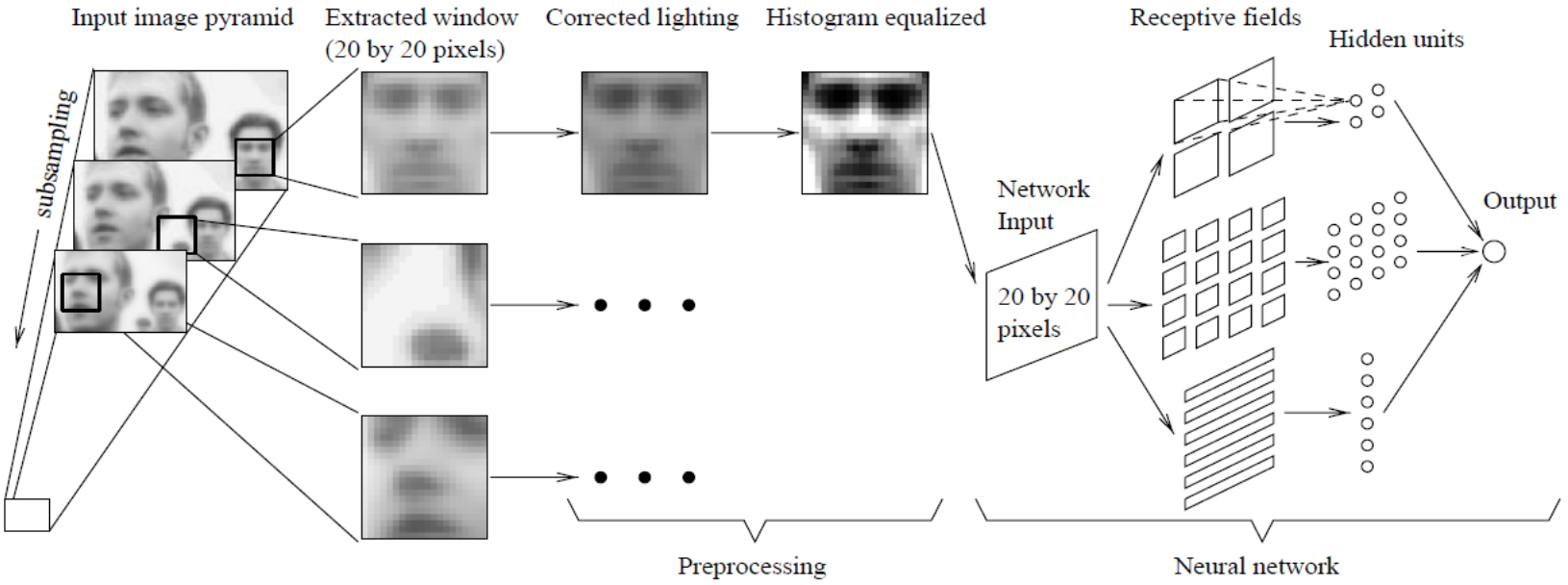


Sliding windows
& Classifier each window



1998

Faces



Neural Network-Based Face Detection, Rowley at al., PAMI 1998

Limits

Numbers worked great, and so did faces…

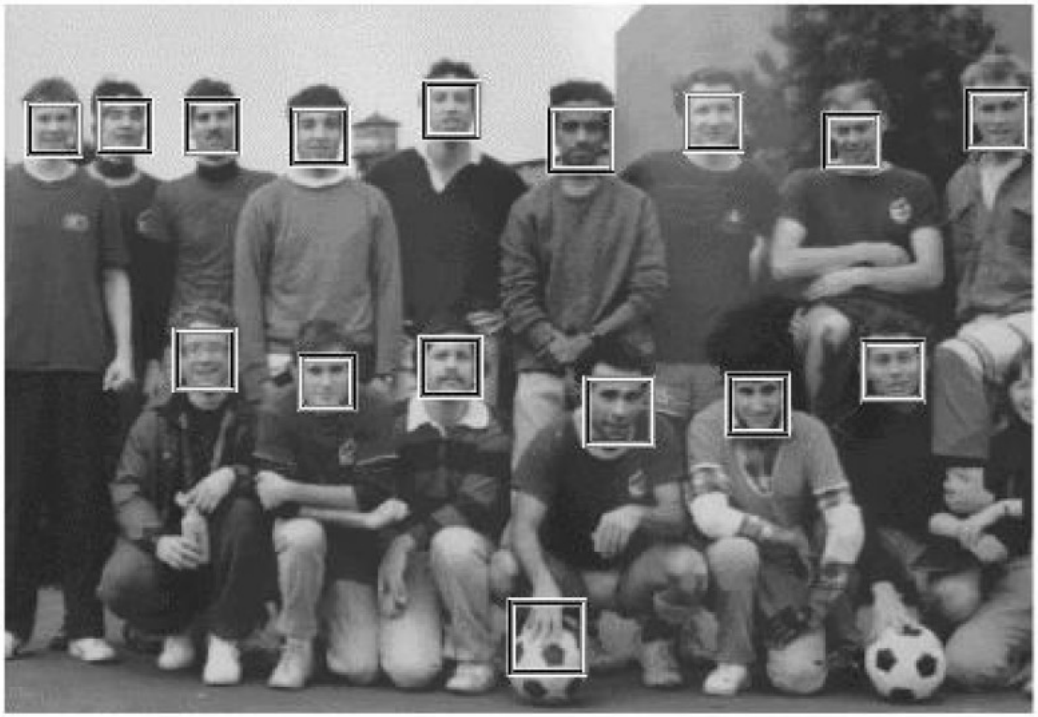
Why did other categories fail?

2 main reasons… (found in 2012)

2001

Sliding window in real time!

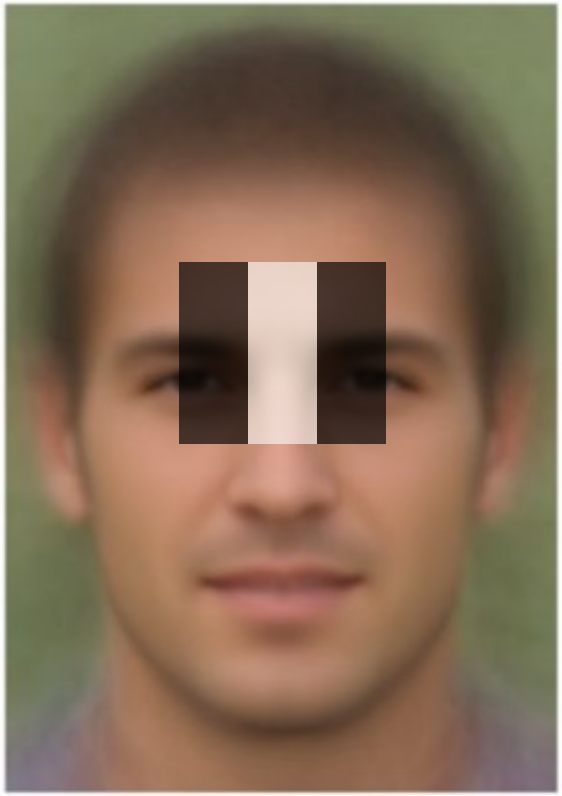
Boosting + Cascade = Speed



Rapid Object Detection using a Boosted Cascade of Simple Features, Viola and Jones, CVPR 2001

Why did it work?





Why did it fail?









1999-2010: Features + Classifier



Ideal Feature Extractor → Ideal Decision Function →

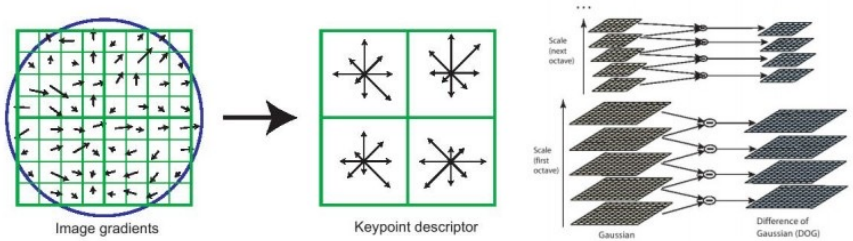
window, top-left
clock, top-middle
shelf, left
…

drawing, middle
hat, bottom right

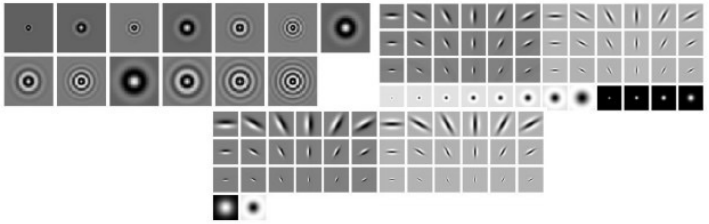
Q: What types of features shall we consider?

Q: What types of decision functions shall we consider?

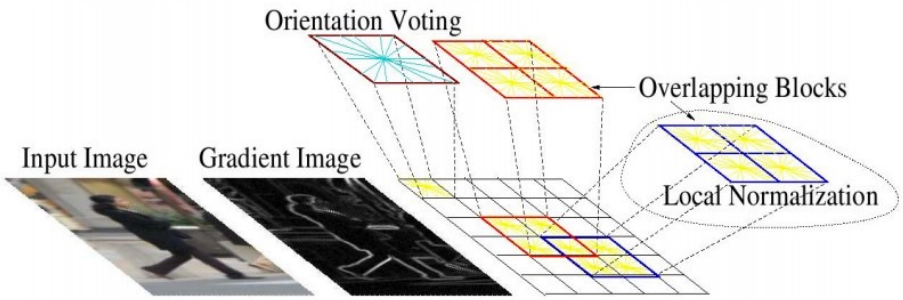
Computer vision features



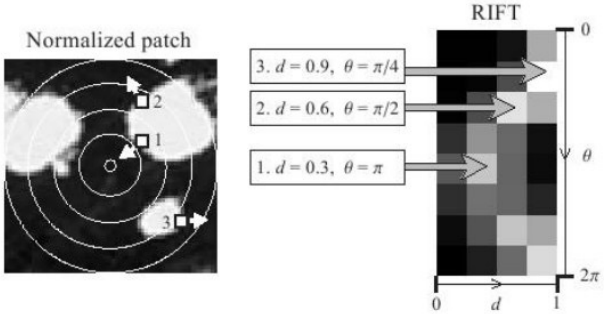
SIFT



Textons

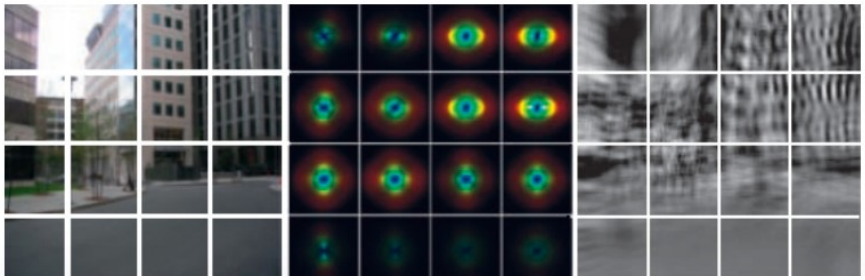


HoG



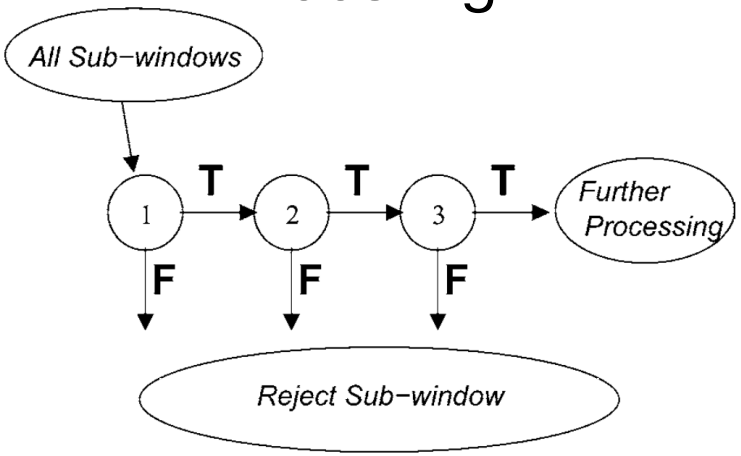
RIFT

GIST

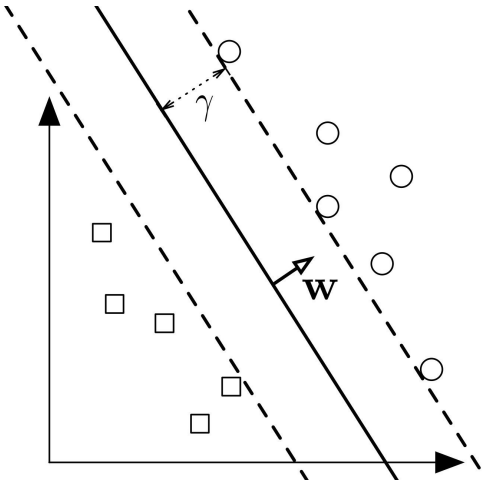


Computer vision decision functions

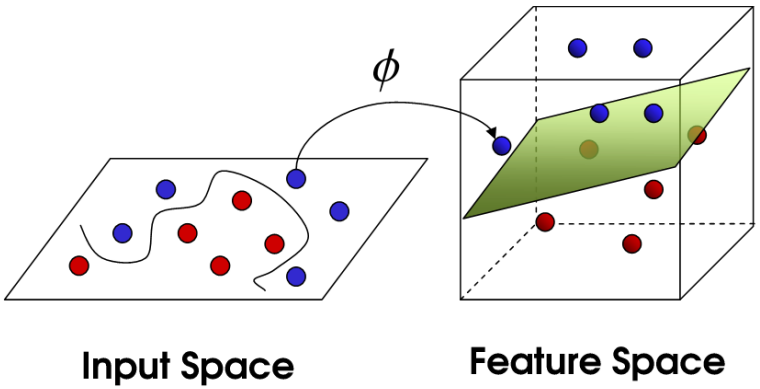
Boosting



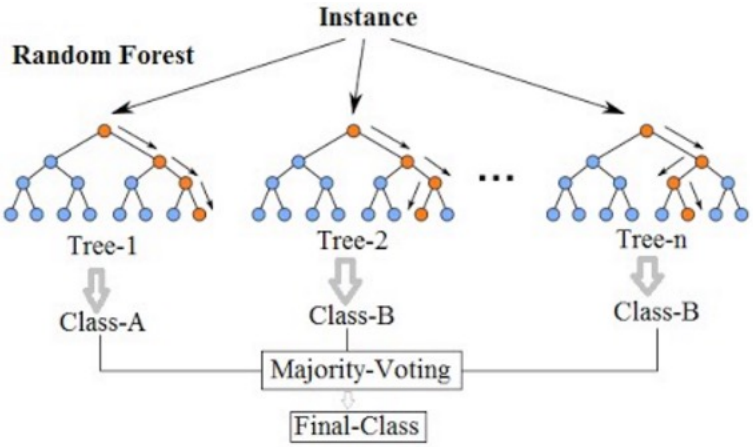
Linear SVM



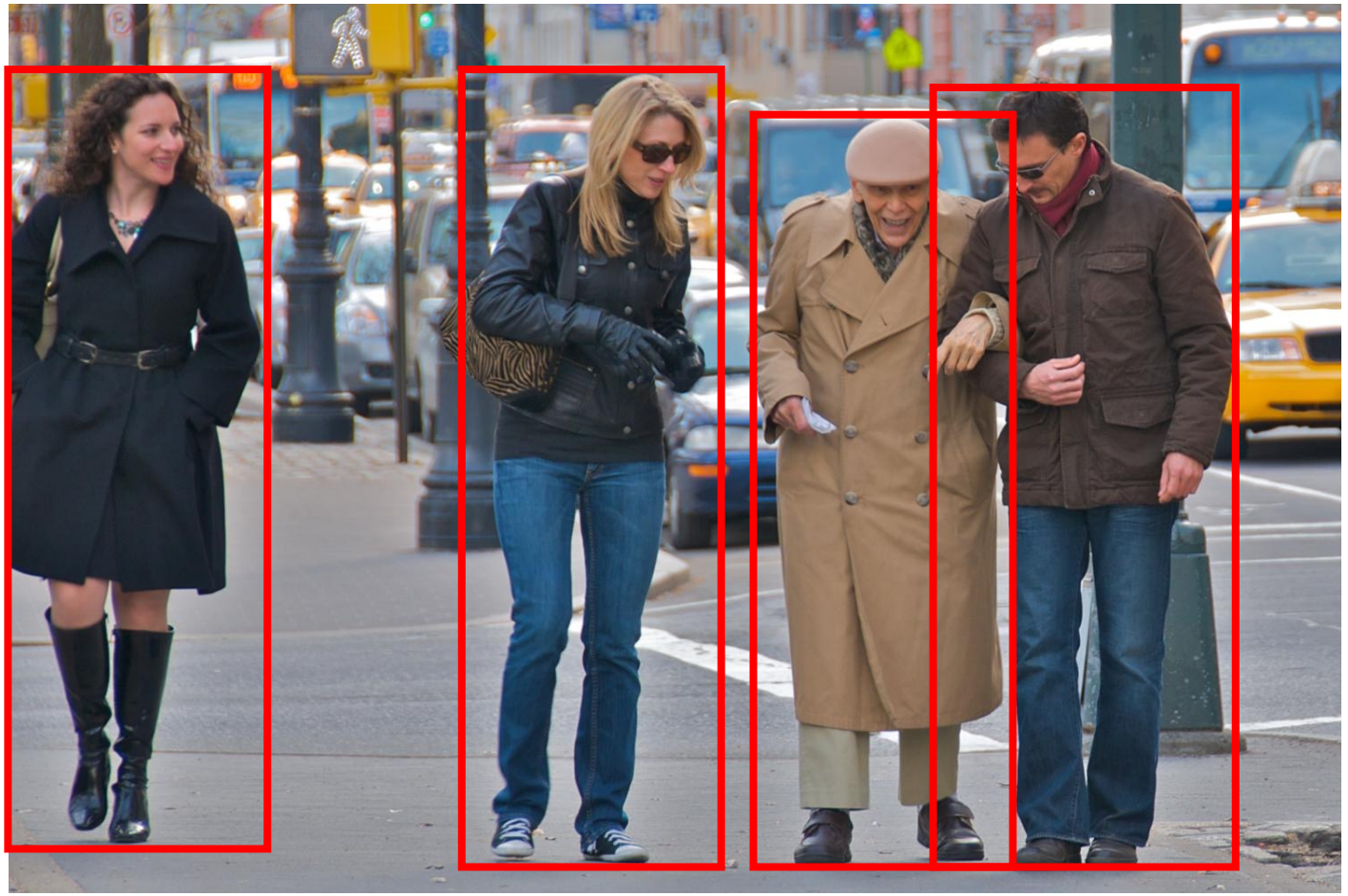
Kernel Machines



Random Forest

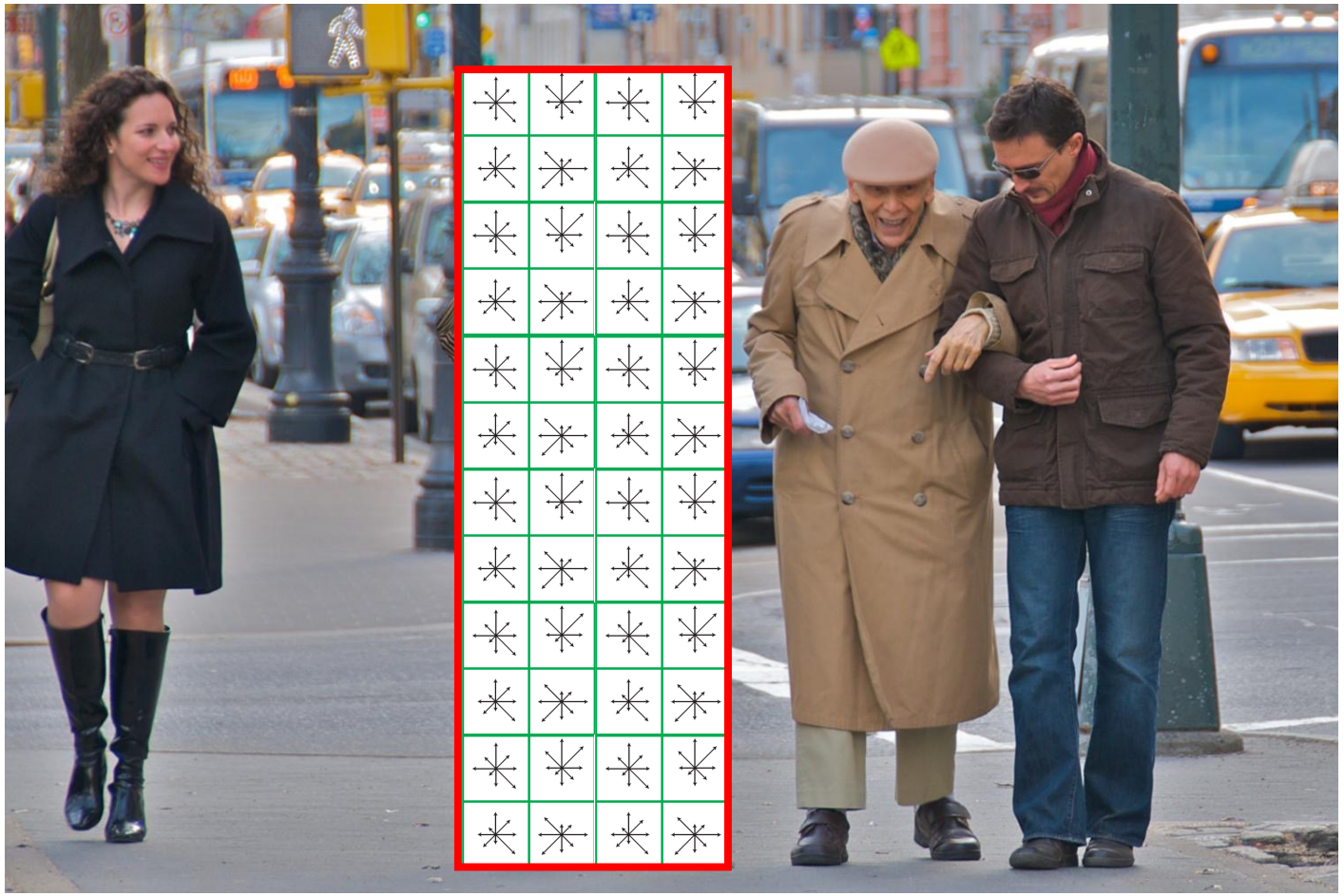


2005 HOG (histograms of oriented gradients)



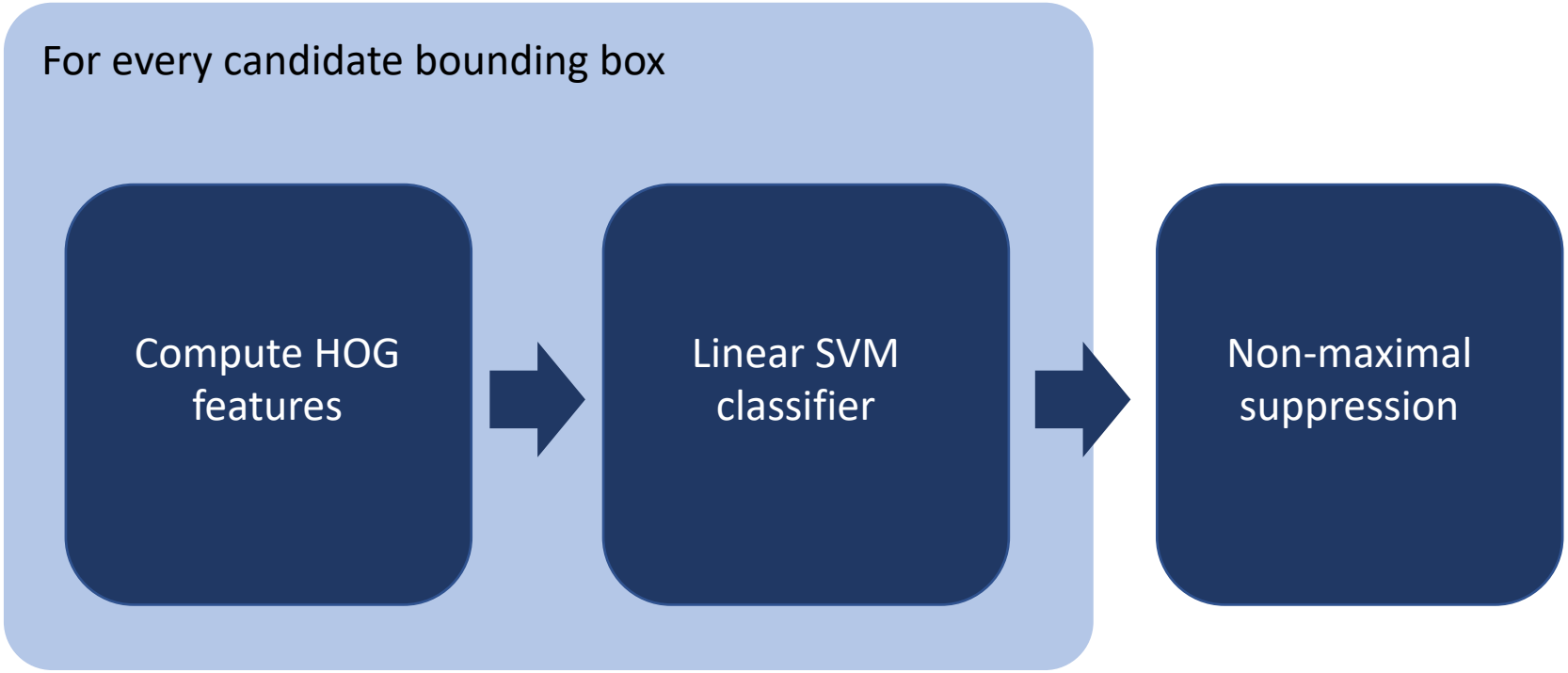
Histograms of oriented gradients for human detection, Dalal and Triggs, CVPR 2005.

2005 HOG (histograms of oriented gradients)



Histograms of oriented gradients for human detection, Dalal and Triggs, *CVPR* 2005.

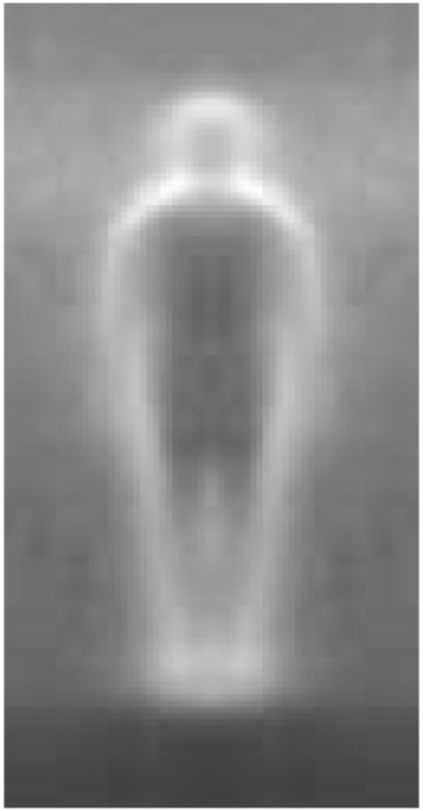
2005 HOG (histograms of oriented gradients)



Why it worked

- Even noisy boundaries give us good cues about the objects
- Hard negative mining
- Computers are fast enough

Why it failed



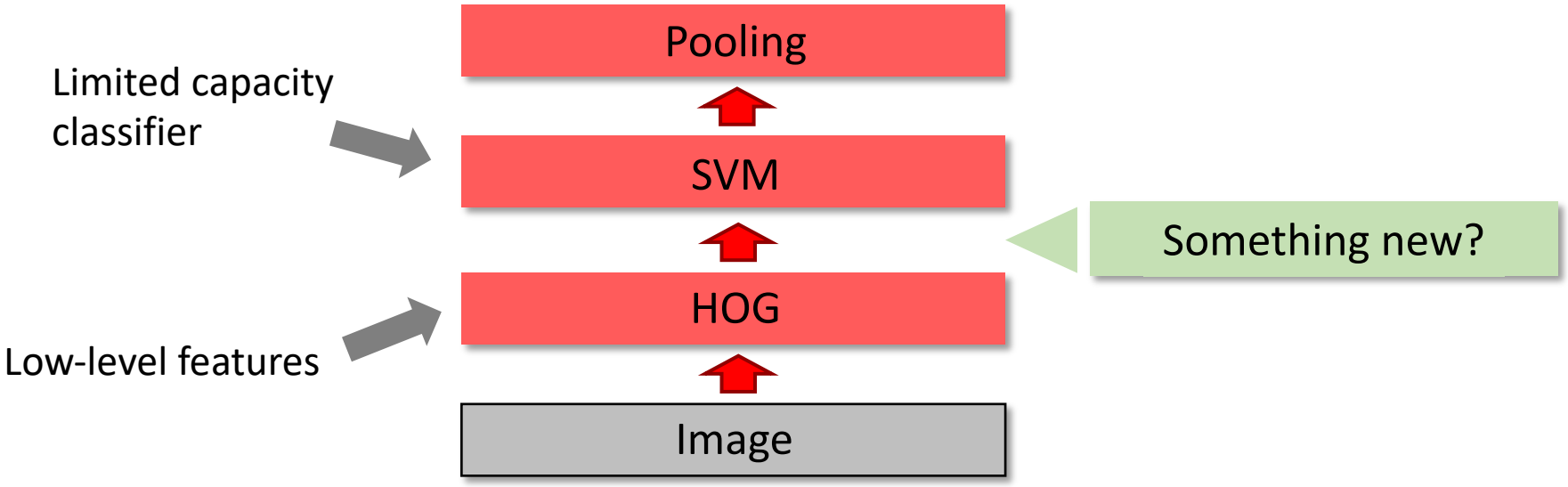








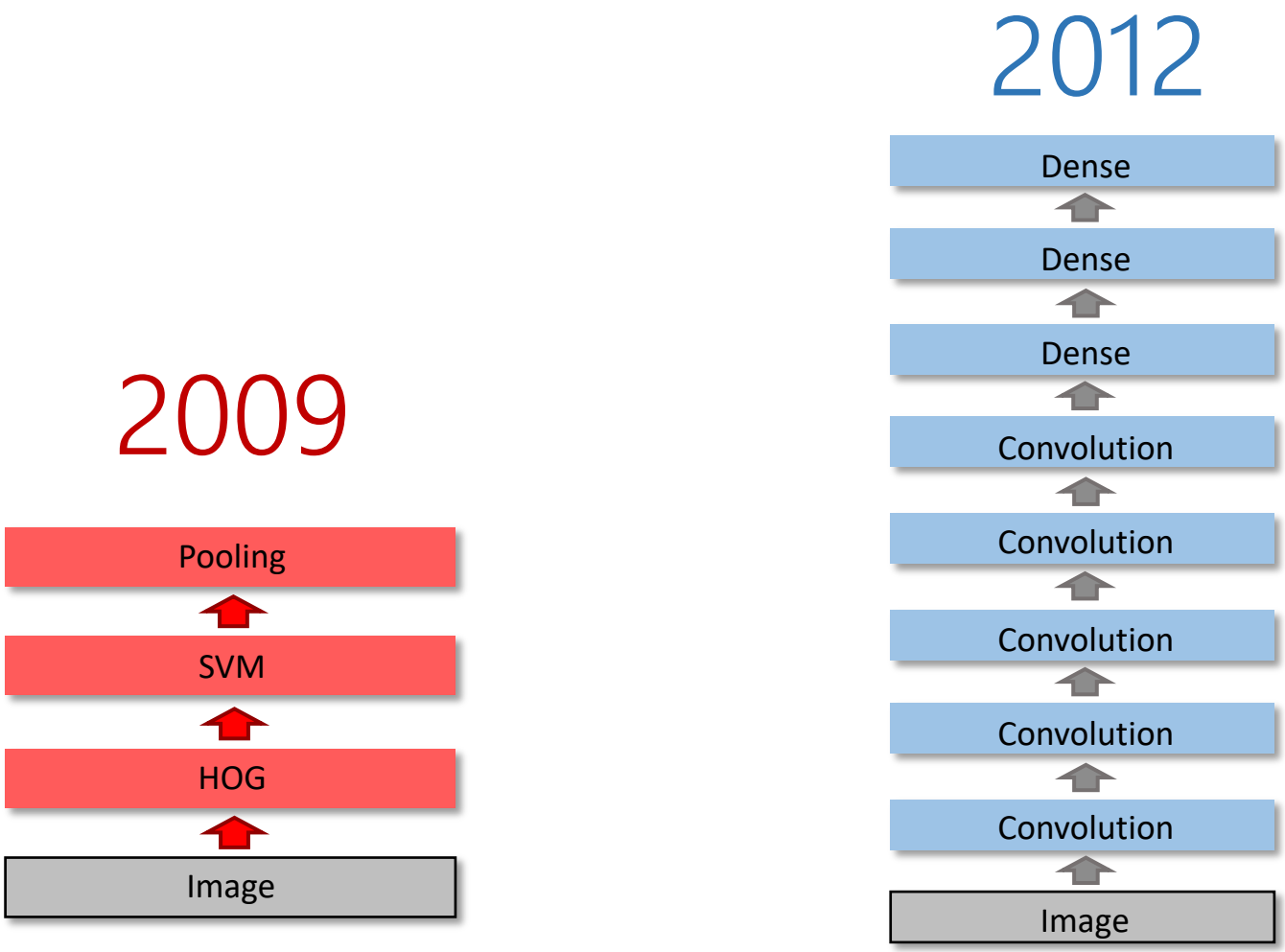
HOG + SVM


Pooling
Limited capacity classifier
SVM
Something new?
HOG
Low-level features
Image

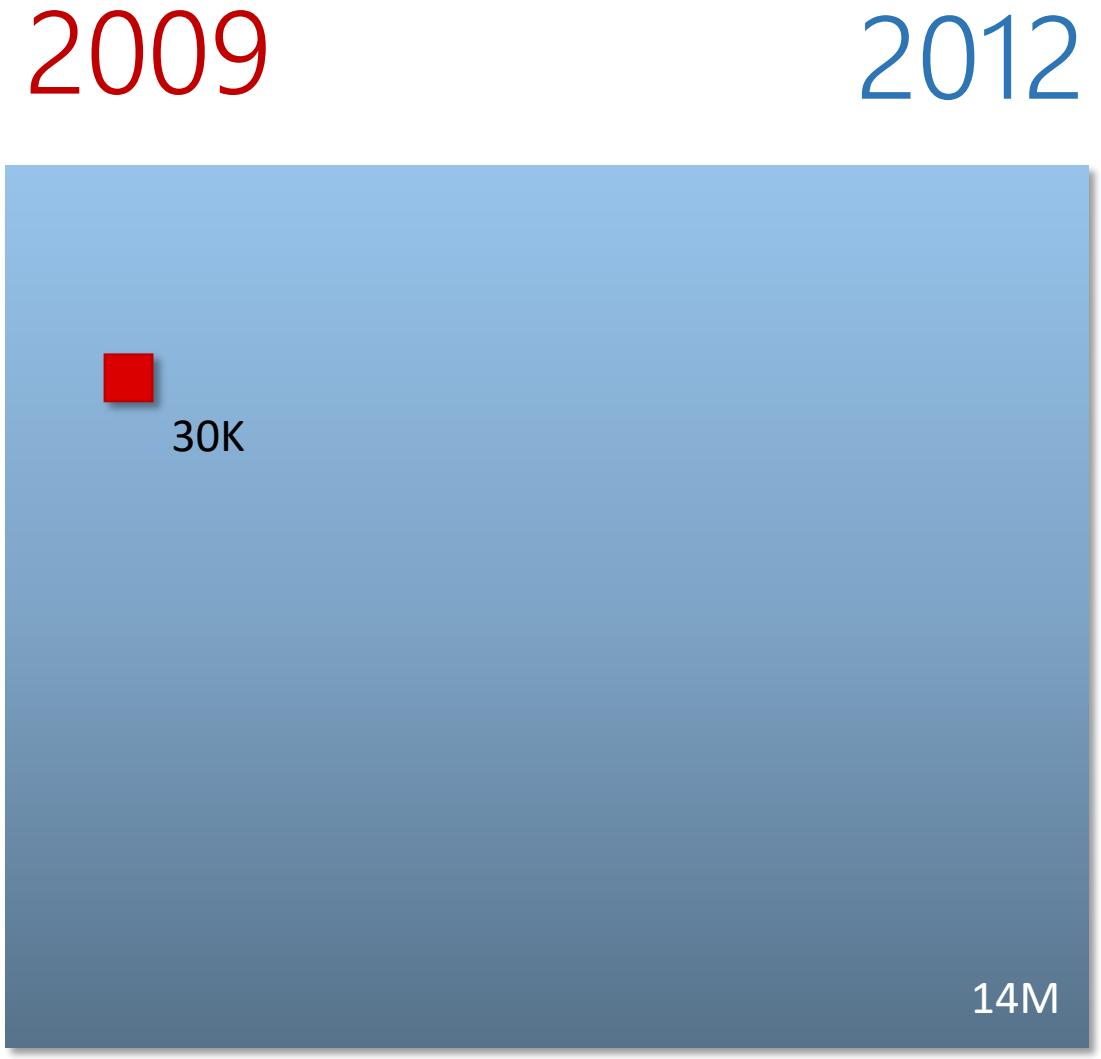
Why deep learning?

deep learning for visual recognition

Algorithms for Learning Visual Features



Images

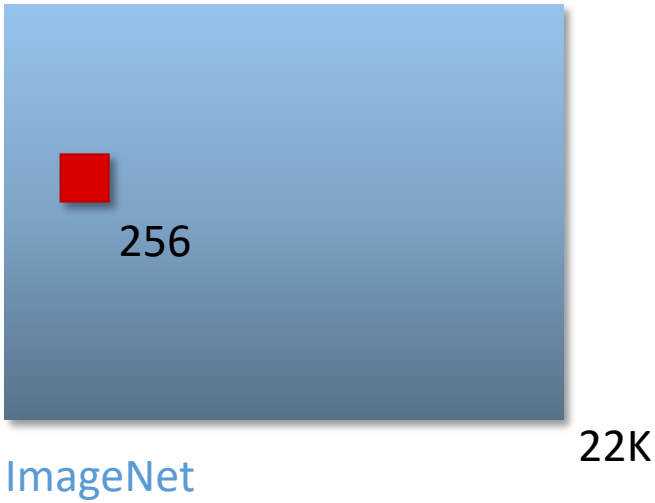


ImageNet

Categories

2009

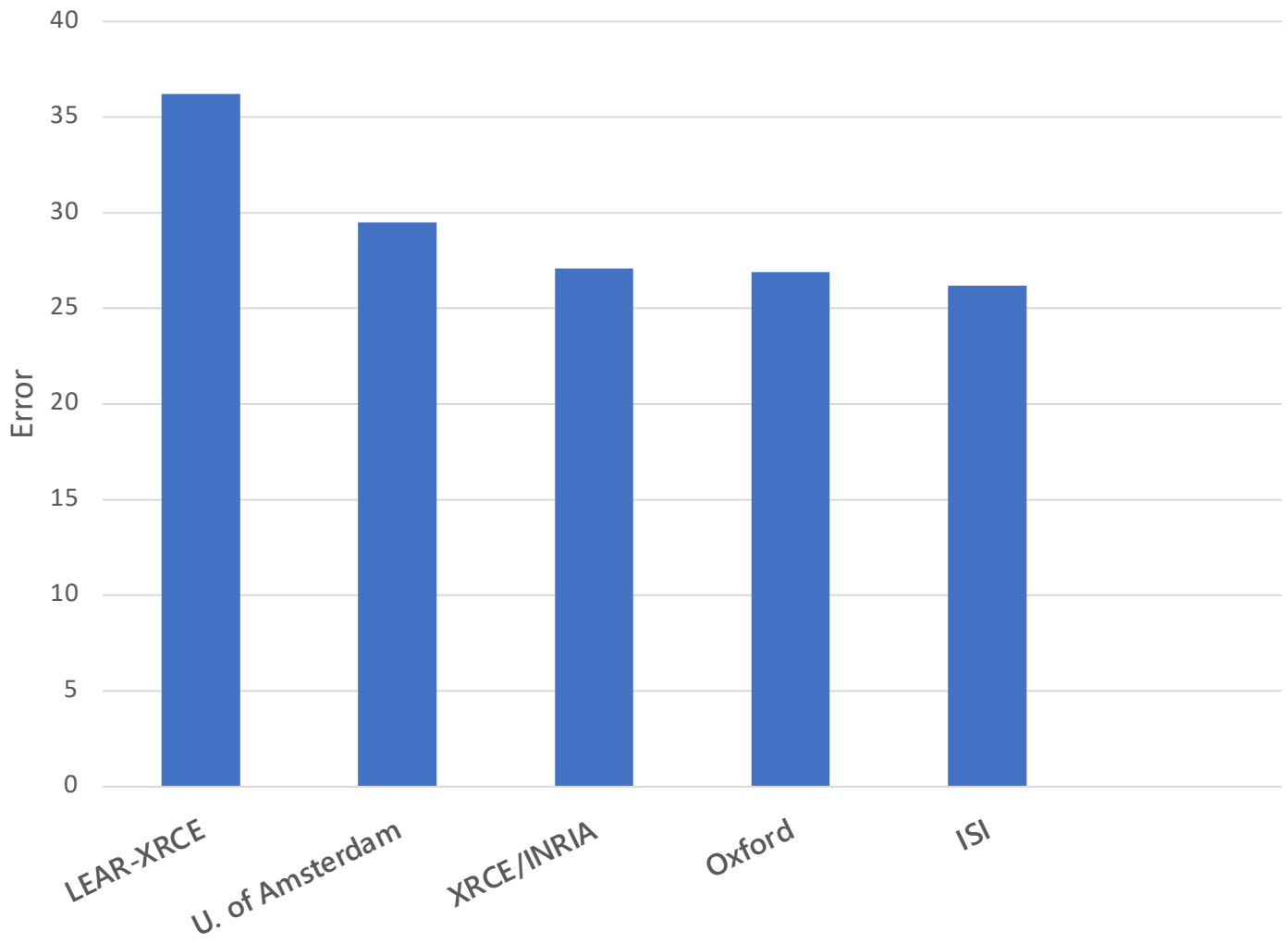
2012



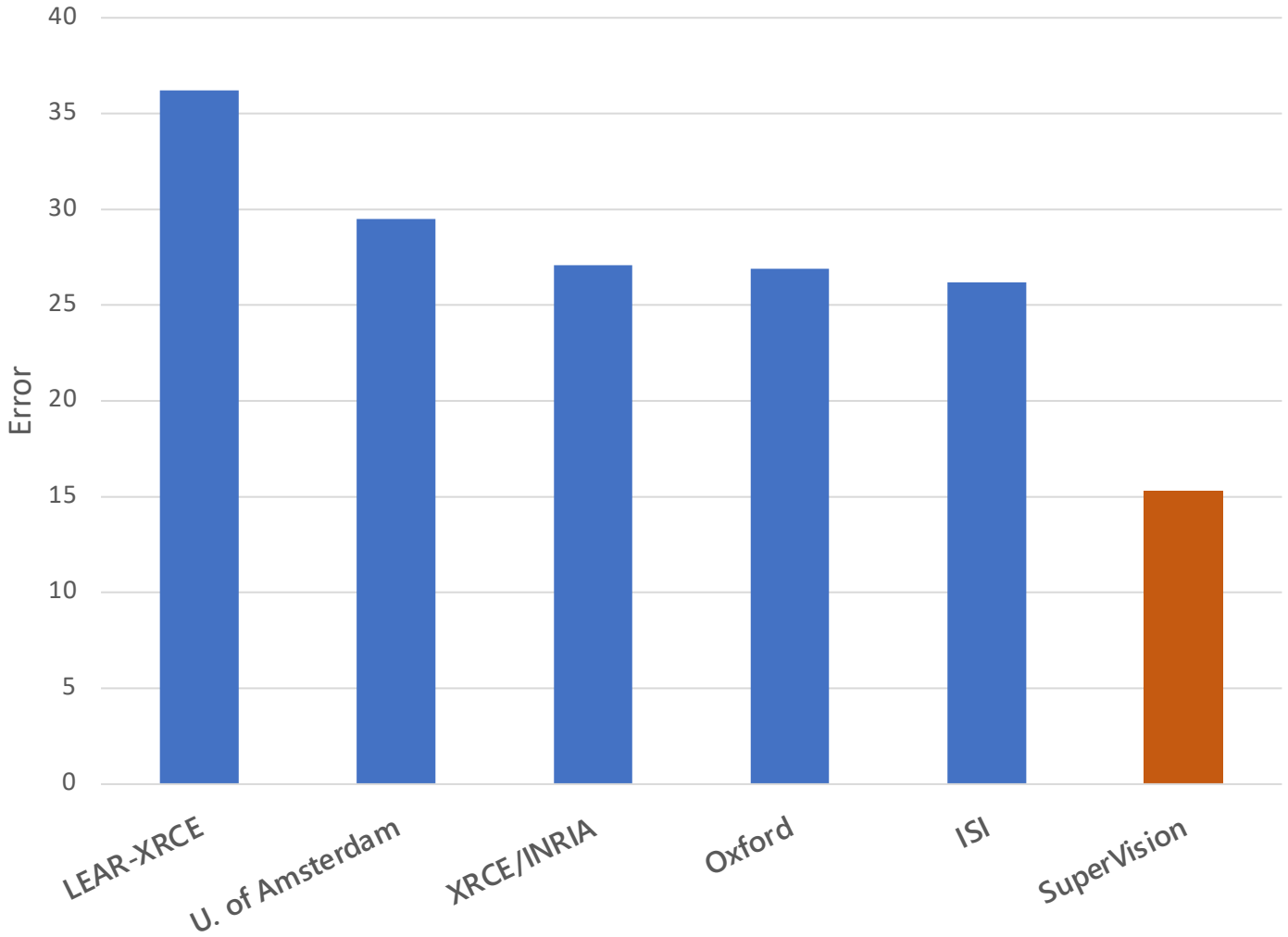
Data
GPUs
\+ Neural Network

?

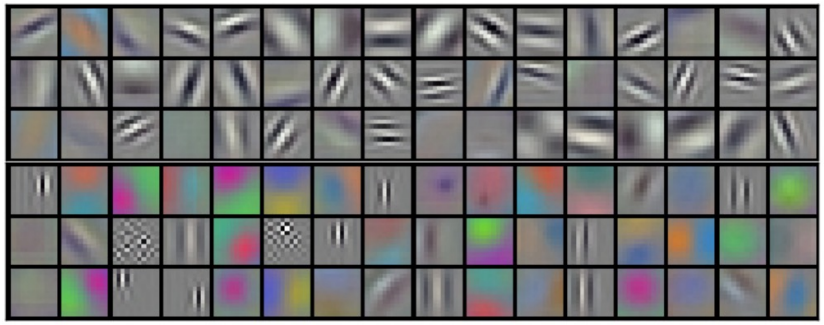
2012 ImageNet 1K (Fall 2012)

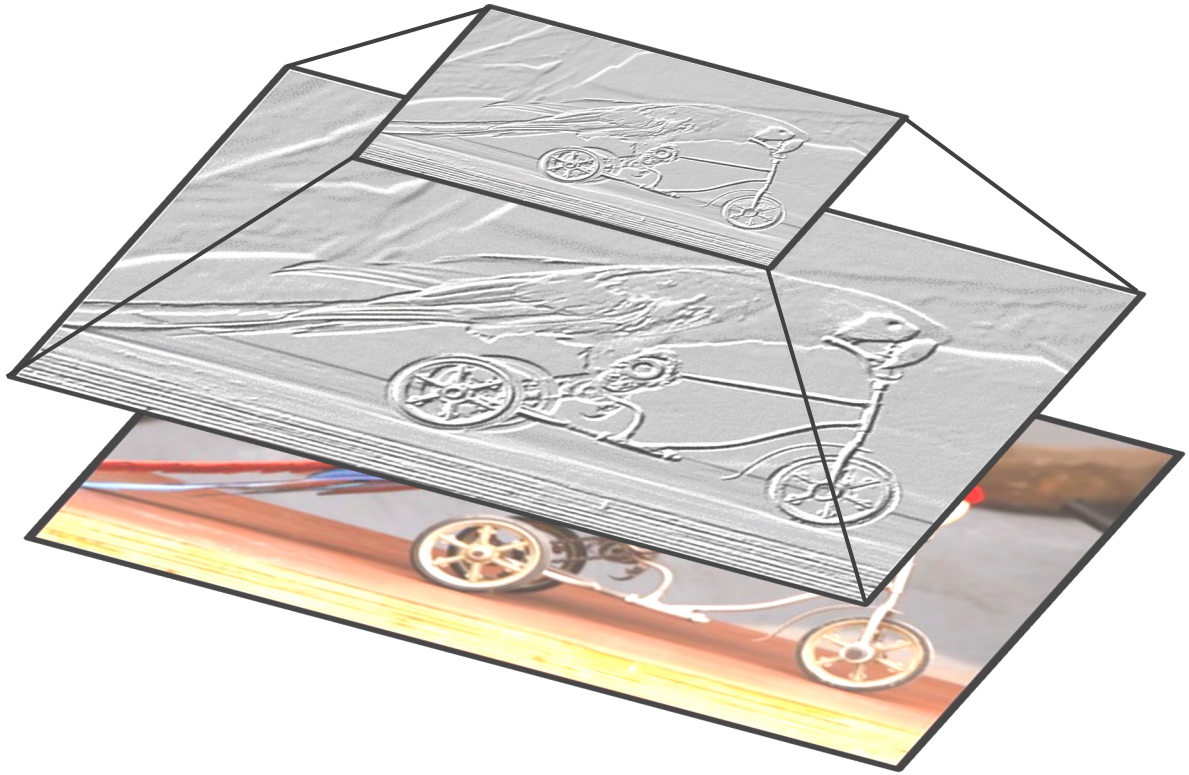


2012 ImageNet 1K (Fall 2012)



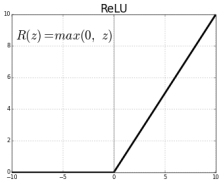
2012





Pooling

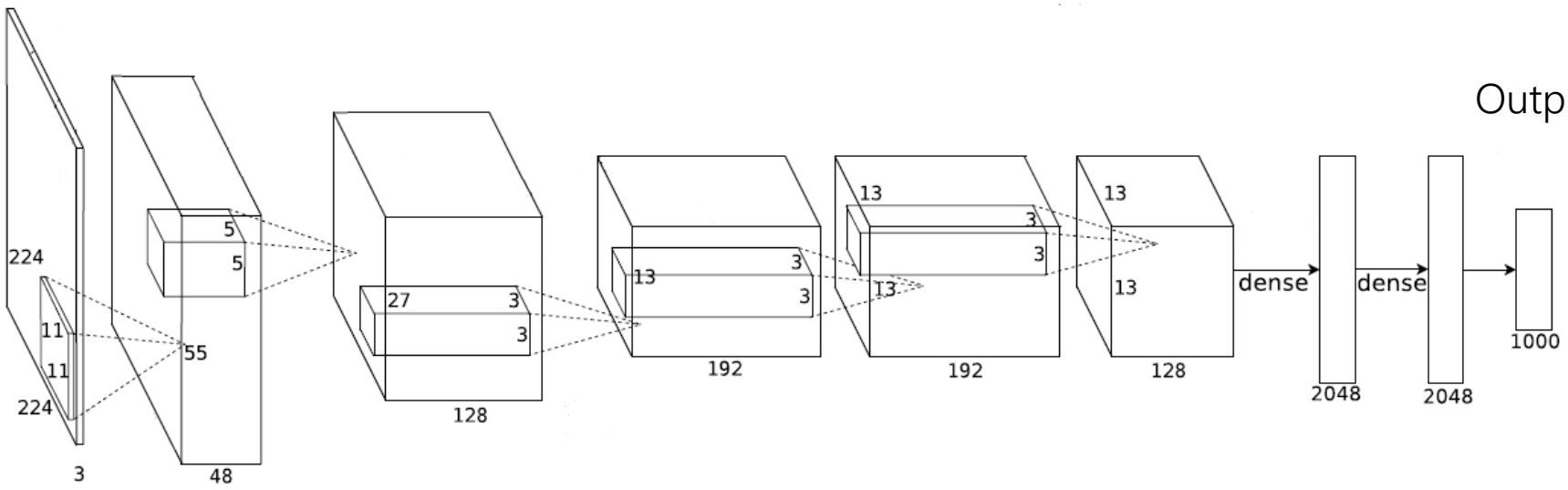
Convolution
+ ReLU



Image

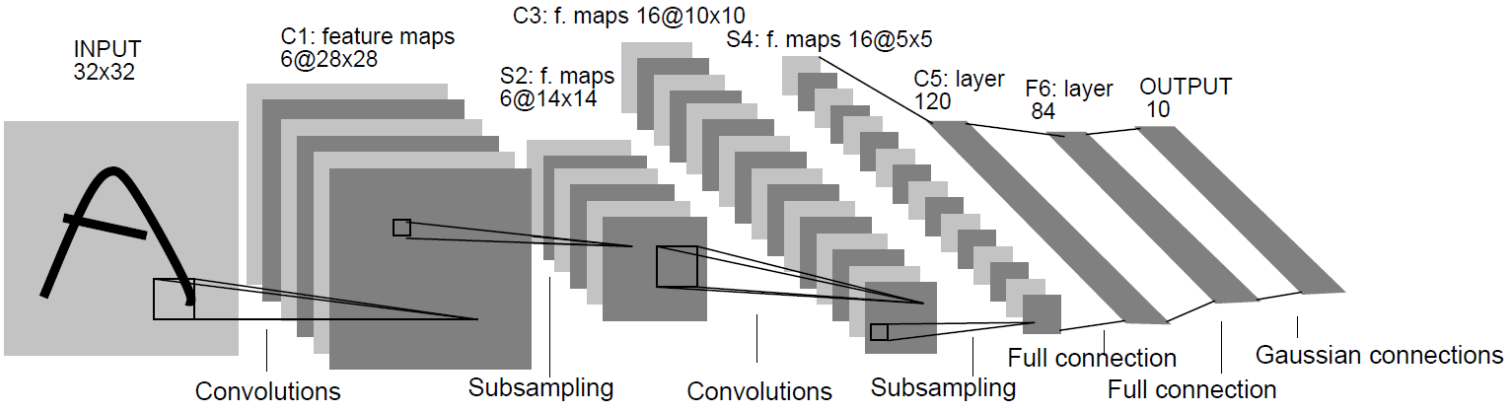
2012 DNNs (deep neural networks)

Image

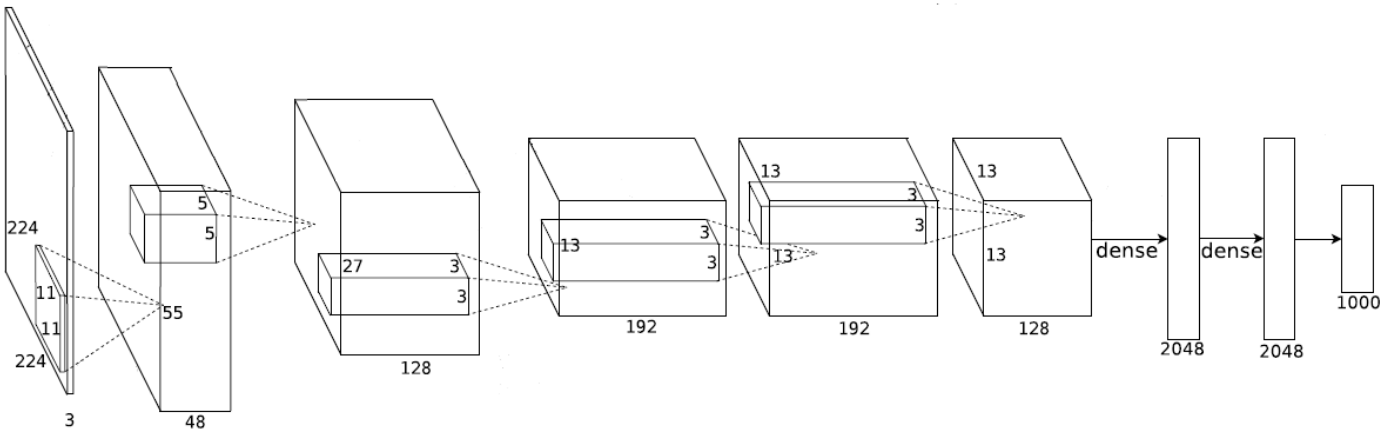


Output

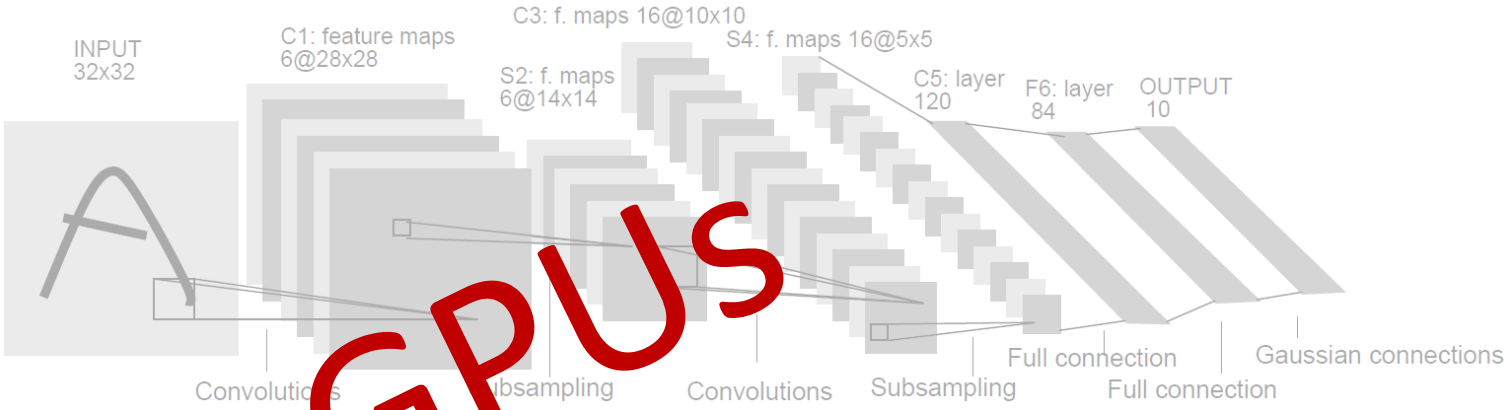
Imagenet Classification with Deep Convolutional Neural Networks, Krizhevsky, Sutskever, and Hinton, NIPS 2012



Gradient-Based Learning Applied to Document Recognition,
LeCun, Bottou, Bengio and Haffner, Proc. of the IEEE, **1998**

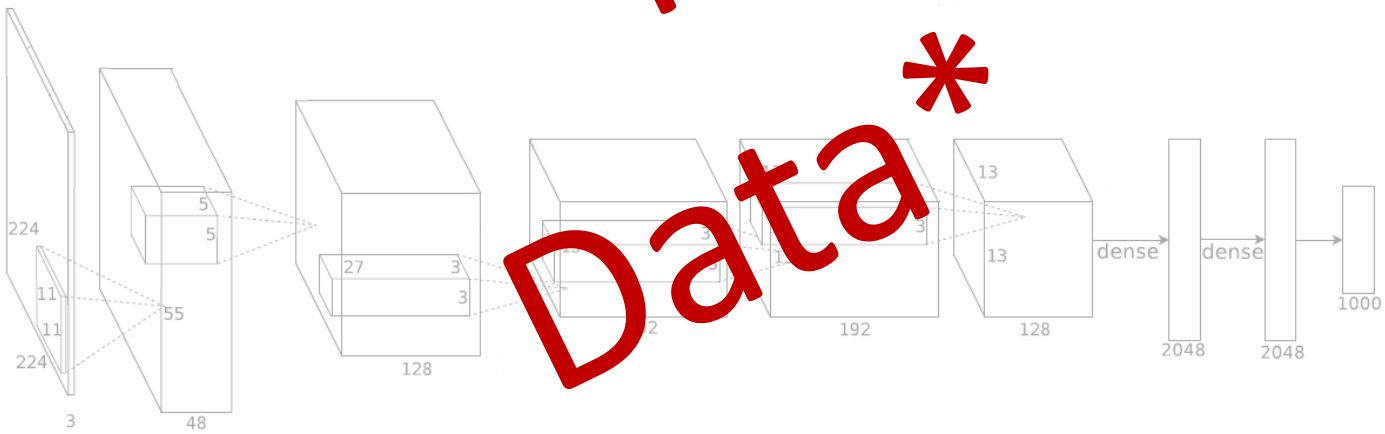


Imagenet Classification with Deep Convolutional Neural Networks,
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Gradient-Based Learning Applied to Document Recognition,
LeCun, Bottou, Bengio and Haffner, Proc. of the IEEE, **1998**

GPUs + Data*



Imagenet Classificati
Krizhevsky, Sutskever

* Rectified activations and dropout

2012 ~2015

Year 2010

NEC-UIUC



Dense descriptor grid: HOG, LBP

Coding: local coordinate, super-vector

Pooling, SPM

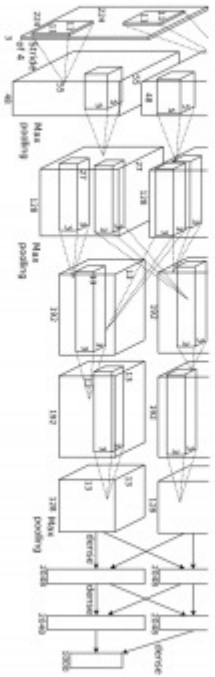
Linear SVM

[Lin CVPR 2011]

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Year 2012

SuperVision

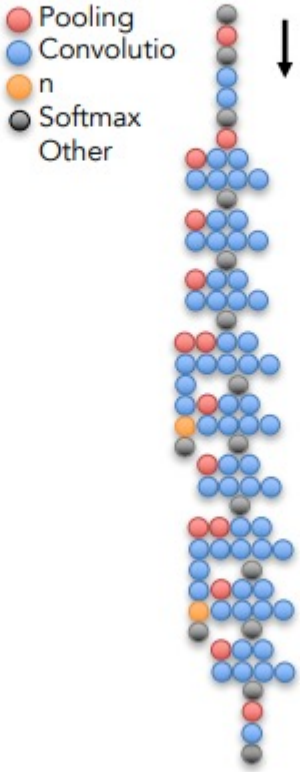


[Krizhevsky NIPS 2012]

Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

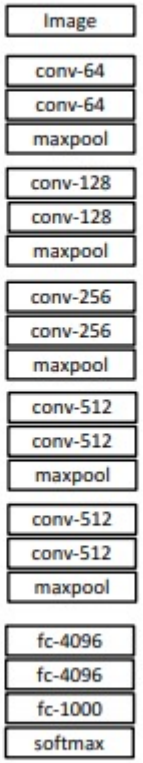
Year 2014

GoogLeNet



[Szegedy arxiv 2014]

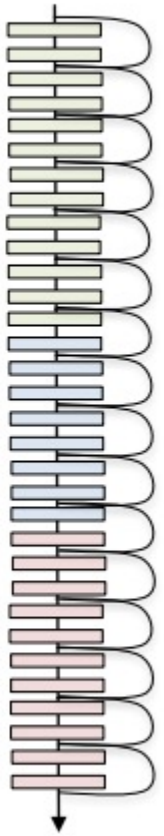
VGG



[Simonyan arxiv 2014]

Year 2015

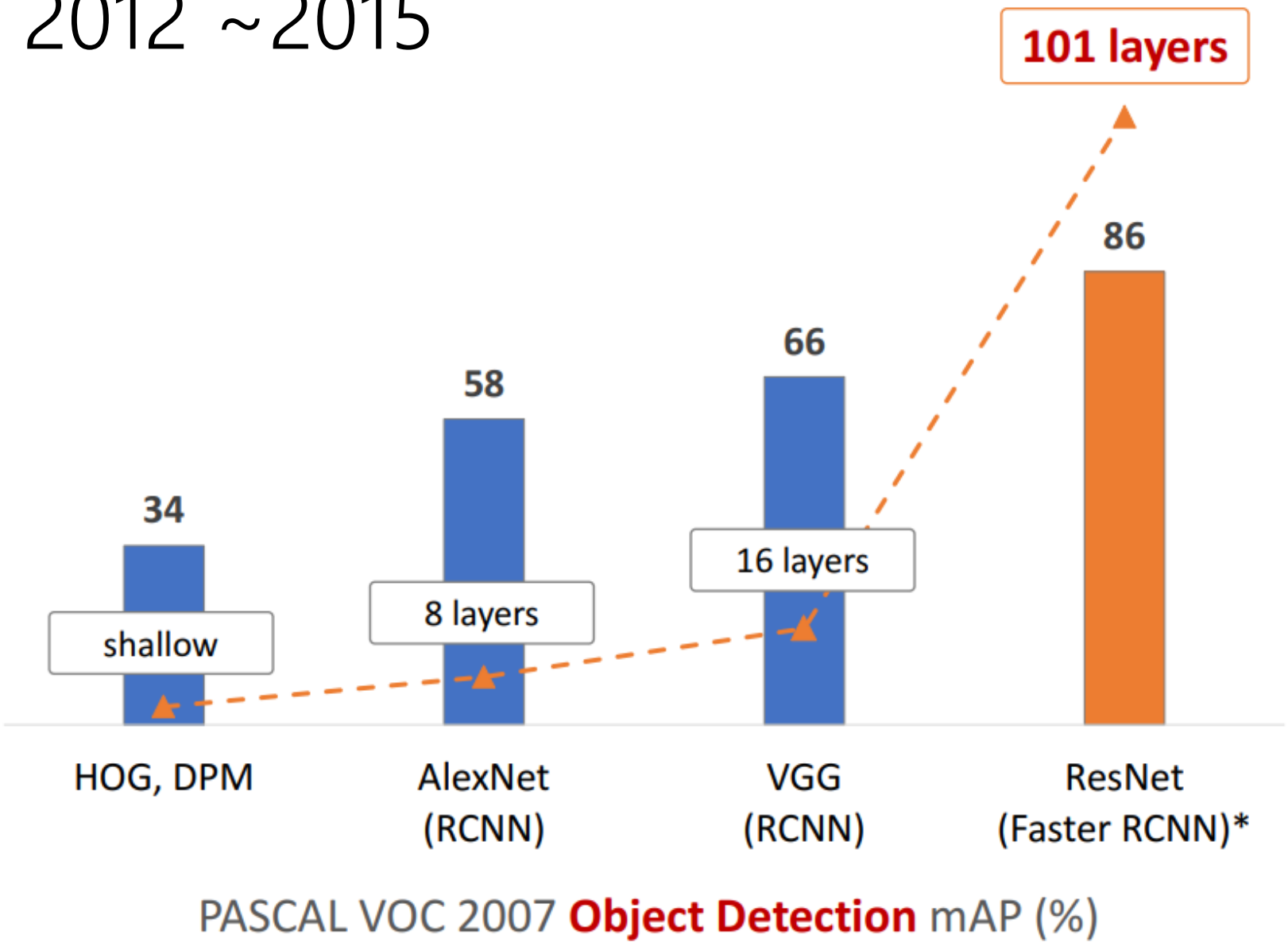
MSRA



[He ICCV 2015]

slide by Fei-Fei, Johnson & Yeung

2012 ~2015



slide by He

2012 ~ 2021

- Architectures
  - AlexNet (ReLU + Dropout)
  - Inception Network
  - Residual Network
- Models
  - CNN / Transformers / RNN
  - GAN / VAE / Diffusion Models
- Initialization & Normalization
  - Better initialization
  - Batch Norm
- Optimization
  - Adam, etc …

- Image Classification
- Object Detection
- Semantic Segmentation
- Image Generation
- 3D Understanding
- Visual Question Answering
- Image Captioning
- Action Recognition
- Medical Image Analysis
- Robotics
- And many many more …

2021 ~ Now

- The rise of vision-language models as foundation models
  - Contrastive language-image pre-training (CLIP)
  - Text condition image generation using diffusion models

- Advances in large language models (LLMs) have significantly accelerated progress in AI, including computer vison
  - Multimodal LLMs that uses text as the central interface for representation learning and output generation

2012 ~ Now



slide by Fei-Fei, Johnson & Yeung

- Data to build observation models
- Data to build priors about the visual world
- Use the models and prior information to infer

Learning based methods for Computer Vision

Less about developing machine learning models
More on using learning models for vision problems!

“What does it mean, to see? The plain man's answer (and Aristotle's, too) would be, to know what is where by looking.”

-- David Marr, *Vision (1982)*

The end